Insurance Inside and Outside the Firm

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Abstract

How much insurance do workers receive against labor market risk inside and outside firms? This paper builds and estimates a new equilibrium search model with optimal wage contracts and assets. In this model, workers use their assets to smooth their consumption, and firms take this into account when they design how wages change with tenure and after shocks. Workers receive significant insurance both inside and outside firms, and the insurance that workers receive outside firms, through financial markets, crowds out the insurance that they receive inside firms, through wage contracts. This has implications for inequality and policy. First, wealthy workers receive more insurance outside firms, but substantially less inside. Second, public insurance policies, such as expansions of household access to credit or unemployment insurance, influence how firms insure workers.

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1 Introduction

What determines the growth and volatility of worker earnings in the labor market? How can workers insure themselves against fluctuations in their earnings? The existing literature has tackled these issues from two distinct perspectives. The first is to consider that firms influence the earnings of workers through the design of optimal wage contracts (Baily, 1974, Azariadis, 1975, Thomas and Worrall, 1988, Burdett and Coles, 2003, Shi, 2009, Menzio and Shi, 2010, 2011, Lentz, 2014, Balke and Lamadon, 2022). From this perspective, workers receive insurance *inside the firm* when they receive wages that are stable over time and in response to shocks. The second is to consider that workers use their own liquid assets to insure themselves (Bewley, 1977, İmrohoroğlu, 1989, Huggett, 1993, Aiyagari, 1994, Kaplan and Violante, 2014). From this perspective, workers receive insurance *outside the firm* by trading risk-free bonds subject to borrowing constraints. Existing empirical evidence shows that these two views are linked: workers at the top of the wealth distribution subsequently experience higher wage growth and pass-through of shocks than workers at the bottom (Guiso, Pistaferri and Schivardi, 2013, Fagereng, Guiso and Pistaferri, 2018, Halvorsen, Ozkan and Salgado, 2022), suggesting that these workers receive different mixes of insurance inside and outside the firm.

Despite decades of research on these issues, the literature has, so far, not brought these two perspectives together. Existing models on optimal wage contracts make the stark assumption that workers have no access to financial markets. By contrast, existing models on precautionary savings make the stark assumption that wage contracts are exogenous. As a result, these models cannot account for existing evidence that workers across the wealth distribution experience different wage profiles. More broadly, we understand very little about the interaction between the insurance that workers receive inside the firm, through wage contracts, and outside the firm, through financial markets.

This paper builds a new equilibrium search model with optimal wage contracts and assets, which brings together the dynamic wage contracting model of Menzio and Shi (2010) with canonical precautionary savings models in the tradition of Bewley (1977). In this model, workers use their assets to smooth their consumption, and firms take this into account when they design how wages change with tenure and in response to productivity shocks. I characterize the optimal contract and estimate it using French administrative data. My main results are that workers receive significant insurance both inside and outside firms, and that the insurance that workers receive outside firms substantially *crowds out* the insurance that they receive inside firms. This has implications for inequality and policy. First, relatively wealthy workers experience substantially higher wage growth and

volatility, consistent with empirical evidence. These workers, however, receive higher wages on average as a compensation for the risky wage contracts they receive. They also enjoy a more stable consumption than poor workers, meaning that they enjoy more insurance overall despite receiving less insurance from firms. Second, public insurance policies, such as expansions of household access to credit or unemployment insurance, influence how much insurance firms provide to workers.

The model features risk-averse workers who can trade risk-free bonds and face a borrowing constraint. They move between employment and unemployment, and can switch jobs under directed search. Workers receive dynamic wage contracts from risk-neutral firms. These contracts specify how wages change with tenure and in response to firm productivity shocks. Contracts are subject to two sets of frictions. First, the search decision of workers is private information, which leads to moral hazard. Second, workers and firms cannot commit to transfers after employer-to-employer (EE) transitions and to transfers after employment-to-unemployment (EU) transitions. In the baseline model, the savings decision of workers and their initial assets are public information, and thus contractible, but I show in section 5 that my results are robust to relaxing these assumptions.

I first clarify that allowing workers to trade assets influences optimal contracts because it expands the set of feasible contracts, even when assets are public information. To show this, I consider an alternative model in which workers and firms can commit to transfers post EE and EU transitions. In this model, any optimal contract can be implemented with zero asset holdings for workers, so assuming that workers are hand-to-mouth is not restrictive. By contrast, in my model optimal contracts use assets to substitute for these missing transfers. For example, if workers and firms could only commit to transfers post EE transitions, the optimal contract would feature an extreme degree of wage frontloading with workers holding assets to self-insure against unemployment risk. Conversely, if workers and firms could only commit to transfers post EU transitions, the optimal contract would feature an extreme degree of wage backloading with workers using all of their existing assets to pay firms a hiring fee. In my model, no transfer can be implemented so assets substitute for both, leading to a trade-off where assets are used to enhance worker retention and to insure workers against unemployment risk.

This trade-off between worker retention and insurance can be characterized using the optimality conditions of the contract. The first is a consumption growth condition, which is standard in models with hand-to-mouth workers and captures how firms optimize worker retention by manipulating wages (e.g. Balke and Lamadon, 2022). The second is a new pseudo Euler equation showing that assets play two roles. First, they are used to smooth the worker's consumption over time and across states given a path for wages,

as in a standard Euler equation. Second, they are used by firms to influence directly the EE transition rate of workers because of wealth effects and because assets influence the set of contracts available to workers in subsequent jobs. Quantitatively, however, these additional effects are very small and have virtually no influence on the optimal contract. Thus, in practice the optimal contract selects paths for wages and assets that optimize worker retention and satisfy the worker Euler equation.

The model has rich implications for wage tenure profiles and the pass-through of productivity shocks to wages, which are both critical determinants of earnings growth and volatility. For instance, wage tenure profiles can be substantially more backloaded than with hand-to-mouth workers because firms take advantage of workers' ability to smooth their consumption using their existing assets to backload wages even more. Tenure profiles can also be less backloaded, and even frontloaded, if workers' desire to accumulate precautionary savings against unemployment risk is strong. Introducing assets also allows firms to pass productivity shocks through differently to the wage and consumption of workers. In fact, the pass-through to consumption can now be interpreted as the product of the pass-through to wages and of the marginal propensity to consume (MPC) out of earnings shocks. Overall, the amount of insurance that firms provide to workers, measured by the degrees of wage backloading and pass-through, depends on the ability of workers to access insurance outside firms using their assets. This has implications for crowding out that I explore using the estimated model.

I estimate the model using French administrative data to quantify how much insurance workers receive inside and outside the firm. To discipline how much insurance workers receive inside firms, I target the tenure profile of wages, a measure of wage backloading, and the pass-through of productivity shocks to wages estimated as in Guiso, Pistaferri and Schivardi (2005). To discipline how much insurance workers receive outside firms, I target the MPC out of earnings shocks from Blundell, Pistaferri and Preston (2008). In the estimated model, workers receive significant insurance both inside and outside firms. This follows from the facts that the tenure profile, the pass-through, and the MPC are all relatively small in the data.

The model has key predictions for the sources of insurance that workers receive across the wealth distribution. Relative to poor workers, wealthy workers enjoy significant insurance outside firms but substantially less insurance inside firms in the sense that they receive wage contracts that are more backloaded and with a higher degree of pass-through to wages. This is consistent with the empirical evidence mentioned before on the heterogeneity in wage growth and volatility across the wealth distribution (Guiso et al., 2013, Fagereng et al., 2018, Halvorsen et al., 2022). In the model, this result stems from

crowding out: firms choose to provide less insurance to wealthy workers precisely because they can smooth their consumption better than poor workers. The crowding out is quantitatively large: workers in the top decile of the asset distribution experience wage growth and pass-through that are about twice as large as those of workers in the bottom decile. However, despite receiving significantly less insurance inside firms, I find that wealthy workers enjoy more insurance overall as their consumption is more stable over time and in response to shocks.

A surprising implication of the model is that relatively wealthy workers also receive higher average wages. The reason is that firms can improve retention more effectively when they provide less wage insurance. Thus, expected firm profits are higher when hiring wealthy workers than when hiring poor workers. In equilibrium with a free entry condition, these higher expected profits lead to higher job finding rates and higher average wages. Despite being relatively modest quantitatively, these differences are critical because they induce wealthy workers to search for contracts that provide less insurance.

The crowding out between insurance inside and outside firms has novel implications for public insurance policies. For instance, an expansion of credit access, modeled as a lower borrowing constraint for workers, improves workers' ability to smooth their consumption and therefore induces firms to offer wage contracts that are more backloaded and with a higher degree of pass-through. Similarly, increasing unemployment benefits induces firms to backload wages more because workers' desire to accumulate precautionary savings falls. Firms also respond by reducing the pass-through of productivity shocks to wages because workers have fewer assets to absorb these shocks. Overall, firms respond to these policies by providing less insurance to workers so they compensate them with higher average wages, but these compensations are quite small quantitatively. By contrast, the crowding out is so large that the consumption growth and pass-through of employed workers barely move. Performing similar exercises in a model with exogenous contracts would have overstated the impact of these policies on consumption.

In the baseline model, I assume that firms can observe the initial assets of newly matched workers and their savings decision and use this information to design wage contracts. These assumptions make the model tractable but it is natural to wonder whether they are critical for my results. I find that they are not. The reason is that workers' incentives to misreport their savings decisions are quantitatively very small so a contract designed under the assumption of hidden savings yields virtually identical outcomes. Besides, workers across the wealth distribution receive different contracts because they choose to search in different markets, and not because firms take advantage of the information they have about their initial assets. In particular, wealthy workers select wage

contracts that are more backloaded because these contracts pay more on average. To establish these results, I rely on the methods of Werning (2001) and Ábrahám and Pavoni (2008) to solve models with hidden savings and Guerrieri, Shimer and Wright (2010) to solve models with directed search and adverse selection.

Related literature There is an important difference between the optimal contract considered in this paper and the contracts studied in the existing literature with savings, such as on unemployment insurance (Kocherlakota 2004, Shimer and Werning 2008) or on hidden effort (Rogerson 1985). In my setting, there are natural restrictions to the contracting space preventing workers and firms from making transfers after the match terminates. As a result, introducing assets expands the contracting space and influences allocations even if assets are observable by firms. By contrast, in the existing literature it is common to assume that firms and workers can make payments conditional on all realizations of observable variables, such as search outcomes or output. For example, in the optimal unemployment insurance literature it is common since Hopenhayn and Nicolini (1997) to assume that the government can tax the income of a worker who successfully found a job, which is equivalent to assuming that transfers post EE transitions can be implemented in my model. This is the reason why, in the existing literature, introducing assets does not influence contracts when they are public information and why previous work has focused on hidden savings. In my model, I find the new effect of assets on the contracting space to be orders of magnitude larger than the effect of hidden savings.

A growing literature has used models with dynamic wage contracts to account for empirical evidence on the sources of earnings risk (Lagakos and Ordoñez 2011, Ábrahám, Álvarez-Parra and Forstner 2013, Balke and Lamadon 2022, Azzalini 2023, Souchier 2023, Agostinelli, Ferraro, Sorrenti and Treuren 2025, Malgieri and Citino 2025, Nattinger and Rothbaum 2025, Acabbi, Alati and Mazzone 2026, Meeuwis, Papanikolaou and Schmidt 2026, Talesara and Wickard 2026). The common thread in this literature is that firms provide insurance to workers. My paper addresses key empirical evidence in support of this idea, namely that workers with different access to financial markets receive different wage contracts. My paper also shows how to relax the assumption that workers are hand-to-mouth, used in this entire literature. Doing this shows that workers are effectively less risk averse against shocks when they can trade assets but this increase in risk tolerance is not uniform across workers: it is much higher at the top of the wealth distribution than at the bottom. I also find that wage contracts influence consumption dynamics even after the job terminates, through their effects on savings.

There also exists a literature bringing together labor market transitions and asset accu-

mulation. An early example is Krusell, Mukoyama and Şahin (2010) who study precautionary savings in a DMP model where wages are set by Nash bargaining. Several articles have introduced assets in search models with EE transitions (Lise, 2013, Chaumont and Shi, 2022, Alves, 2025, Caratelli, 2025, Kaas, Lalé and Siassi, 2026) but in these models wage contracts are subject to ad hoc restrictions (e.g. wages are constant within jobs). My paper instead introduces optimal wage contracts, which is critical to study the sources of insurance that workers receive against labor market risk.

Some of my results on contracts with assets relate to existing work. Stevens (2004) shows that one implementation of optimal contracts with risk-neutral workers is extreme wage backloading. Oikonomou (2018) shows that unemployment risk leads to extreme wage frontloading. My paper also relates to Pissarides (2004) who shows that firms insure workers against unemployment risk by delaying separations and Kiyotaki and Zhang (2024) who show that workers' initial assets influence human capital accumulation when workers cannot commit to stay with the employer that trains them. Ábrahám and Laczó (2018) show that public storage improves allocation in a model where two agents share risk subject to limited commitment.

The paper starts in section 2 with the model, which I characterize in section 3. Section 4 quantifies how much insurance workers receive inside and outside firms and explores the implications of the crowding out for inequality and policy. Finally, section 5 revisits the assumption that assets are observable. Proofs are in the appendix.

2 A model with wage contracts and assets

I first present an equilibrium model of the labor market in which workers transition between employment and unemployment and across jobs subject to search frictions, receive dynamic wage contracts from firms and trade assets to smooth their consumption. This model replaces the assumption that workers are hand-to-mouth in Menzio and Shi (2010) by the assumption that workers trade risk-free bonds subject to borrowing constraints, in the spirit of Bewley (1977). I also introduce exogenous EE transitions, also known as godfather shocks, for the quantitative analysis.

2.1 Environment

Time is discrete and runs forever.

Workers and firms A continuum of ex-ante homogeneous workers can be employed or unemployed. Workers receive a wage when employed, and home production b when unemployed. They have utility $u(c)$ over consumption with discount factor β.

Firms are owned by foreign diversified investors, so they are effectively risk-neutral with discount rate r. An active firm is one that is matched with a single worker. The output from that match x_t follows the mean reverting process

$$\log x_t = \rho \log x_{t-1} + \sigma \nu_t$$

where ν_t are i.i.d. innovations with a standard normal distribution. ρ is the persistence of productivity and σ its volatility.

Financial markets This is a small open economy with exogenous interest rate r. Workers can save using risk-free bonds a_{t+1} subject to the borrowing constraint

$$a_{t+1} \geq 0$$

Labor market transitions Workers transition from unemployment to employment (UE), across jobs (EE) and from employment to unemployment (EU). UE transitions are endogenous. Workers can make two types of EE transitions. First, they can make *endogenous* EE transitions by searching for a job. Second, workers can make *exogenous* EE transitions when they receive outside job offers with probability δ^{ee}. During exogenous EE transitions, the poaching firm offers workers the continuation value they would have received at their current job[1]. EU transitions are exogenous and occur with probability δ[2].

Timing Each period, the sequence of events is as follows:

a) productivity shocks ν_t are realized,

b) exogenous EE and EU transitions: separations into unemployment occur and workers receive outside job offers at random,

[1]One concern is that poachers might not be willing to offer this value to workers during exogenous EE transitions if the worker continuation value is too high. In model simulations, I find that this concern is valid for less than 5% of exogenous EE transitions, and among these transitions the losses are very small. For these reasons, I do not model the problem of poachers during exogenous EE transitions explicitly.

[2]To keep the exposition simple, I do not allow for endogenous quits into unemployment. However, I verify ex-post that this restriction does not bind in the quantitative model. The reason why it does not bind is that workers can quit with EE transitions by searching for a job with a high job finding rate.

c) endogenous UE and EE transitions: employed workers who did not receive exogenous job offers and workers unemployed at the start of the period search for jobs; firms post vacancies; matching occurs,

d) firms produce and pay current wages; workers make saving decisions and consume.

Directed search with assets Endogenous UE and EE transitions are modeled with directed search. There is a continuum of labor markets indexed by the promised value to a worker denoted v and the assets of workers a. Every period, workers choose in which labor market to search and firms choose where to post vacancies. Both employed and unemployed workers search in the same labor markets. I assume that workers with assets a can only search in markets indexed by a, and not elsewhere, but I revisit this assumption in section 5.

Denote $\phi_u(v,a)$ and $\phi_e(v,a)$ the mass of unemployed and employed workers searching for a job and $\phi_f(v,a)$ the mass of vacancies posted by firms. Let κ denote the search intensity of employed workers relative to unemployed workers. In each labor market, a constant returns to scale matching function $\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f)$ turns workers searching for a job and vacancies into matches. Define the job finding rate $\tilde{\lambda}_w(\phi_u + \kappa\phi_e, \phi_f)$ as the probability that an unemployed worker finds a job, and the vacancy filling rate $\tilde{\lambda}_f(\phi_u + \kappa\phi_e, \phi_f)$ as the number of workers matched per vacancy. These rates are defined in the usual way as

$$\tilde{\lambda}_w(\phi_u + \kappa\phi_e, \phi_f) \equiv \frac{\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f)}{\phi_u + \kappa\phi_e}, \quad \tilde{\lambda}_f(\phi_u + \kappa\phi_e, \phi_f) \equiv \frac{\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f)}{\phi_f}$$

Since these matching rates will depend on v and a in equilibrium, we can write them in short-hand notation as

$$\lambda_w(v,a) \equiv \tilde{\lambda}_w(\phi_u(v,a) + \kappa\phi_e(v,a), \phi_f(v,a)), \quad \lambda_f(v,a) \equiv \tilde{\lambda}_f(\phi_u(v,a) + \kappa\phi_e(v,a), \phi_f(v,a))$$

The job finding rate for employed workers is $\kappa\lambda_w(v,a)$.

Unemployed workers Unemployed workers receive an endowment b, choose how much to save and consume and in which labor market v to search. Given the equilibrium job

finding rate, $\lambda_w(v,a)$, the value of unemployed workers at the end of the period satisfies

$$\begin{aligned} U(a_t) = \max_{c_t, a_{t+1}, v_{t+1}} \quad & u(c_t) + \beta \left[\lambda_w(v_{t+1}, a_{t+1}) v_{t+1} + (1 - \lambda_w(v_{t+1}, a_{t+1})) U(a_{t+1}) \right] \\ \text{s.t.} \quad & c_t \leq (1+r)a_t + b - a_{t+1} \\ & a_{t+1} \geq 0 \end{aligned}$$

Appendix A.4 shows that the choice of savings a_{t+1} follows a standard Euler equation where workers smooth their consumption by depleting their savings over time. The choice of search v_{t+1} follows a standard trade-off: searching in a high-v labor market brings a higher value v conditional on a match, but these matches occur with lower probability because $\lambda_w(v,a)$ decreases with the value v in equilibrium.

Contracts Wage contracts specify wages w_t for each history of shocks and conditional on the worker's initial asset position a_0 and on delivering initial value v to workers[3]. The key contracting friction in this model is that the worker's search decision is private information, which leads to moral hazard. I assume that the savings decision of workers a_{t+1} is public information, and therefore contractible, but I relax this assumption in section 5. I also assume that firms do not respond to counteroffers[4].

A complete contract would also specify transfers conditional on EE or EU transitions. I rule them out by assuming that workers and firms cannot commit to these transfers, which implies that they must be equal to 0 after any history in the optimal contract. This assumption will turn out to be critical, so I will return to it later.

2.2 Optimal wage contracts

Following previous work on dynamic contracts, I write the contract recursively in terms of promised values and continuation values instead of histories of shocks. Denote V_t the promised value of an employed worker at the start of the period. The state of a match at the beginning of the period, denoted $s_t \equiv \{V_t, a_t, x_t\}$, is the worker's promised value V_t, the asset of the worker a_t, and the firm productivity x_t.

The components of the contract at time t are the wage paid today, the savings decision of the worker and a set of continuation values for each state tomorrow. Formally, these

[3] I find that allowing for randomization makes virtually no difference numerically in my model. For this reason, I focus here on deterministic contracts to keep the notation light.

[4] This assumption can be formally justified as follows: counteroffers are private information to workers, and expire before workers can return to their current employers to negotiate higher wages. Under these conditions, it is optimal for current employers not to respond to outside offers. This assumption also arises naturally in models with endogenous search intensity and where firms seek to discourage workers from searching by not matching counteroffers (Postel-Vinay and Robin, 2004).

components are represented by the functions

$$w_t\left(s_t\right), \quad a_{t+1}\left(s_t\right), \quad V_{t+1}(s_t, x_{t+1})$$

A *contract* is a collection of these functions for all t. In the recursive formulation below, I write the components of the contract as $w_t, a_{t+1}, V_{t+1}(x_{t+1})$ without explicitly mentioning state variables. It will also be convenient to define the continuation value of workers at the current job as

$$W_t \equiv u(c_t) + \beta \mathbb{E}_{x_{t+1}}\left[V_{t+1}(x_{t+1})|x_t\right] \tag{1}$$

where consumption c_t satisfies the budget constraint $c_t + a_{t+1} = (1+r)a_t + w_t$.

Worker value Given the contract, the worker chooses a search strategy to maximize the present value of utility. The value of a worker satisfies

$$V_t = \delta U(a_t) + (1-\delta)(\delta^{ee} W_t + (1-\delta^{ee}) \max_{v_t} \left[\kappa \lambda_w(v_t, a_t) v_t + (1 - \kappa\lambda_w(v_t, a_t))\, W_t\right]) \tag{2}$$

In equation (2), the first term is the continuation value of a worker who becomes unemployed at time t with assets a_t. The second term is the continuation value of a worker who makes an exogenous EE transition and receives the value W_t she would have received at her current job. The third term depends on the probability that a worker finds another job $\kappa\lambda_w(v_t, a_t)$. A worker who makes an endogenous EE transition receives value v_t whereas a worker who does not find another new job receives the value W_t.

Equation (2) shows how assets interact with the contracting frictions. Notice that moral hazard arises because the worker's search decision v_t only depends on the surplus that workers get from EE separations $v_t - W_t$, and not on the firm value. Thus, relative to the search policy that the worker chooses, a firm with a positive value would prefer the worker to search instead in markets with a higher value v_t and a lower job finding rate $\lambda_w(v_t, a_t)$ because it wants to retain the worker. Since the worker search decision v_t is private information, the firm cannot control it directly and instead influences the worker's decision indirectly by manipulating the continuation value at the current job W_t and the job finding rate $\lambda_w(v_t, a_t)$. How can the firm manipulate the job finding rate? The firm can manipulate the worker's assets a_t, which influences the choice of v and thus the job finding rate $\lambda_w(v_t, a_t)$. However, assets also influence the continuation value of workers who become unemployed $U(a_t)$. This dual role for assets generates a trade-off that I describe in detail in section 3.

Optimal contracts Denote the optimal search policy $v(W_t, a_t)$, the implied endogenous EE probability as

$$p_t \equiv p(W_t, a_t) \equiv \kappa \lambda_w(v(W_t, a_t), a_t)$$

and the worker's expected surplus from endogenous EE separations as

$$S_t \equiv S(W_t, a_t) \equiv \kappa \lambda_w(v(W_t, a_t), a_t) \left(v(W_t, a_t) - W_t\right)$$

Finally, denote $\Pi(s_t)$ the present value of profits for a firm matched with a worker who has promised value V_t and assets a_t when productivity is x_t. Taking as given the value of unemployment $U(a)$, the equilibrium job finding rate $\lambda_w(v, a)$, and the search policy of workers $v(W, a)$, the optimal contract solves

$$\Pi(s_t) = \max_{\substack{w_t, c_t, a_{t+1}, \\ V_{t+1}(x_{t+1})}} (1-\delta)(1-\delta^{ee})(1-p(W_t, a_t)) \left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}[\Pi(s_{t+1})|x_t]}{1+r} \right) \tag{3}$$

subject to

$$\begin{aligned} &\text{(PK)}: && V_t \leq \delta U(a_t) + (1-\delta)\left[\delta^{ee} W_t + (1-\delta^{ee})\left(W_t + S(W_t, a_t)\right)\right] \\ &\text{(Budget)}: && c_t + a_{t+1} = (1+r)a_t + w_t \\ &\text{(BC)}: && a_{t+1} \geq 0 \end{aligned}$$

where we recall that $W_t = u(c_t) + \beta \mathbb{E}_{x_{t+1}}\left[V_{t+1}(x_{t+1})|x_t\right]$ and $s_{t+1} \equiv \{V_{t+1}(x_{t+1}), a_{t+1}, x_{t+1}\}$.

The contract maximizes the present value of profits, where $(1-\delta)(1-\delta^{ee})(1-p(W_t, a_t))$ is the probability that the worker remains within the current match this period. The first constraint (PK) is the promise keeping constraint, stating that the value the worker gets from the contract either at the current job, through unemployment or at future jobs must deliver at least the promised value V_t. The second constraint (Budget) is the budget constraint of the worker, and the third constraint (BC) is the borrowing constraint. Firms take as given the optimal search policy of workers $v(W, a)$, so the incentive compatibility constraint for search is implicit in the definition of $p(W, a)$ and $S(W, a)$.

Value of new matches In the first period of employment, firms also solve (3) except that there are no separations ($\delta = \delta^{ee} = p_t = S_t = 0$) and productivity is 1 ($x_t = 1$). Denote the firm value in the first period by $\Pi_0(V_t, a_t)$.

Worker value with transfer I have omitted transfers conditional on separations in the optimal contract (3) because the lack of commitment of firms and workers constrains them

to be 0. However, it is useful to also write the worker value with transfers because section 3.1 discusses the relation between transfers and assets. Denote the transfers after EU transitions τ_t^{eu} and the transfers after endogenous EE transitions τ_t^{ee}. The worker value with transfers is

$$V_t = \delta U(a_t + \tau_t^{eu}) + (1-\delta)\left[\delta^{ee} W_t + (1-\delta^{ee})\left(W_t + S(W_t, a_t + \tau_t^{ee})\right)\right]$$

The key observation is that transfers and assets a_t appear in the same places in this equation so firms can provide insurance and incentives to workers with either. Proposition 1 formalizes this intuition.

2.3 Equilibrium

Free entry Firms post vacancies in each labor market subject to a standard free entry condition. Firms choose in which market (v, a) to post vacancies and the unit cost of posting a vacancy is k. The free entry condition is

$$-k + \lambda_f(v, a)\Pi_0(v, a) \leq 0 \tag{4}$$

with equality for each active market (v, a).

Definition of an equilibrium An equilibrium is a set of value functions, policies and matching rates for each labor market (v, a) such that i) the unemployed worker policies maximize the unemployment value, ii) the firm and employed worker policies satisfy the optimal contract, iii) the free entry condition is satisfied and iv) the job finding and vacancy filling rates are consistent with the matching function. The laws of motion for the distributions $D^u(a)$ and $D^e(s)$ are defined as usual.

2.4 Discussion of assumptions

The most consequential assumption is that workers and firms cannot commit to transfers post transitions τ_t^{ee}, τ_t^{eu}. In fact, section 3.1 shows that letting workers trade assets influences the optimal allocation precisely because these transfers cannot be implemented in the optimal contract. Assuming that transfers post EE transitions τ_t^{ee} cannot be implemented is standard in the literature on optimal wage contracts and is usually justified on the ground that bonded labor is prohibited by law (e.g. Stevens, 2004). Contracts in which workers are forced to repay firms for their investment when they leave are tightly regu-

lated (e.g. Training Repayment Agreement Provisions in the United States) and sometimes difficult to enforce because workers can just default. Assuming that transfers post EU transitions cannot be implemented captures the idea that some risk cannot be insured by firms and thus requires insurance outside firms in the form of precautionary savings. In the case of unemployment risk, firms may not pay severance to workers if the separation occurs because of firm bankruptcy. Besides, even formal commitments about severance pay can be subject to interpretation as they only apply to separations that occur for reasons outside of the worker's control. More broadly, this assumption is a stand-in for other forms of risk that cannot be insured by firms, such as shocks to health expenses or preference shocks inducing workers to quit into unemployment[5].

The assumption that the worker's initial assets and savings decision are observable by firms seems like a strong and unrealistic restriction on optimal contracts. However, section 5 shows that this assumption is fairly inconsequential by solving a model in which assets are private information to workers. This model is much harder to solve and characterize so I use the version with observable assets for my analysis.

The assumption that workers also switch jobs due to exogenous EE transitions is useful for the quantitative analysis. It is a standard feature of dynamic search models (e.g. Jolivet, Postel-Vinay and Robin 2006) and captures the idea that many EE transitions are driven by factors that firms cannot influence, such as the relocation of a spouse. This feature of the model turns out to be useful to match the level of the pass-through of productivity shocks to wages observed in the data. Appendix B.3 provides an alternative version of the model without exogenous EE transitions and with persistent measurement errors instead, and shows that this version also matches the data and has similar quantitative implications. For this reason, I do not see exogenous EE transitions as essential.

3 The role of assets in wage contracts

I now characterize the optimal wage contract by describing how wages, assets and consumption change with tenure and in response to productivity shocks. I first clarify that assets influence optimal contracts because they substitute for missing transfers post EE and EU transitions. Because firms cannot implement those transfers, they use workers' assets to optimize worker retention further but also to insure them against unemployment risk. I then use the optimality conditions of the optimal contract, including a new pseudo Euler equation that I derive, to show how the paths of wages, assets and consumption

[5] Cole and Kocherlakota (2001) show that firms cannot provide state-contingent insurance against income and expense shocks that are private information to workers.

depend on this new trade-off between worker retention and insurance.

3.1 Assets substitute for missing transfers

This section clarifies that assets influence optimal contracts because they substitute for missing transfers post EE and EU transitions. I state the main result in proposition 1 and then explain it using figure 1 for illustration.

Proposition 1. *1. Assume that firms and workers can commit to all transfers* $\{\tau_t^{eu}, \tau_t^{ee}\}$*. Also assume that* $\delta^{ee} = 0$*. Then, the paths of consumption and EE probability* $\{c_t, p_t\}$ *are identical in the optimal contract that satisfies the hand-to-mouth constraint* $a_{t+1} = 0$ *and in the relaxed optimal contract where that constraint is not imposed.*

2. Assume instead that firms and workers can commit to all transfers but one, denoted τ_t^i *where* $i \in \{eu, ee\}$*, so* $\tau_t^i = 0$ *for all* t*. Also assume that* $\delta^{ee} = 0$*, productivity is constant and the borrowing constraint is sufficiently relaxed*[6]*. Then, the paths of consumption and EE probability are identical to those in part 1 and the optimal path for assets satisfies*

$$a_{t+1} = (\tau_{t+1}^i)^* \quad \forall t$$

where $(\tau_{t+1}^i)^*$ *denotes the optimal transfer in the contract in which firms and workers can commit to all transfers and in which the hand-to-mouth constraint is satisfied.*

Proof. See appendix A.1. □

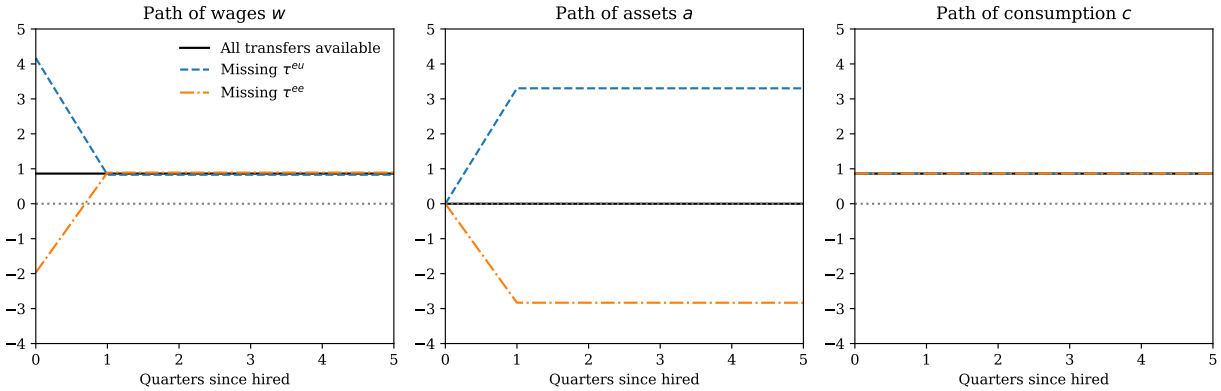
Part 1 from proposition 1 states that the optimal allocations c_t and p_t are identical whether workers are restricted to hold no asset during employment or can trade risk-free bonds, provided that firms and workers can commit to transfers $\{\tau_t^{eu}, \tau_t^{ee}\}$. In this case, letting workers trade assets is therefore irrelevant when assets are public information. This is a well-known result in the optimal unemployment insurance literature: the firm can deliver value to workers either with promises of future wages and transfers or with assets (e.g. see Werning, 2002). Critically, this result requires that transfers $\{\tau_t^{eu}, \tau_t^{ee}\}$ can be chosen freely in the optimal contract. This assumption is in fact standard in the literature on optimal unemployment insurance[7], but not in the literature on optimal wage con-

[6] More precisely, the borrowing limit has to be finite to ensure transversality but large enough so that assets can substitute for transfers freely, even if negative.

[7] Since Hopenhayn and Nicolini (1997), it is common practice in this literature to assume that the government (the principal) can tax workers (the agent) when they find a job. This assumption is equivalent to assuming that workers can commit to transfers when they find a job, that is τ_t^{ee} can be freely chosen. Furthermore, in these models there is no exogenous separation so τ_t^{eu} is irrelevant.

tracts[8]. This is the reason why letting workers trade assets influences optimal wage contracts even when these assets are public information whereas allowing such trades does not influence optimal unemployment insurance contracts. The solid black line from figure 1 shows the implementation of the optimal contract that satisfies the hand-to-mouth constraint $a_{t+1} = 0$ for all t. In this specific example, the worker is matched from unemployment with zero initial asset, productivity is constant, $\delta^{ee} = 0$ and $\beta(1+r) = 1$. The optimal contract implements a wage that is constant over time.

Figure 1: Assets substitute for missing transfers post EE and EU transitions



Note: paths of wages, assets and consumption for an employed worker hired from unemployment at $t = 0$ with zero initial asset. The solid black line shows the model where all transfers τ_t^{eu}, τ_t^{ee} can be implemented, and where the optimal contract imposes $a_{t+1} = 0$ without loss of generality. The blue and orange lines show the models where τ_t^{eu} or τ_t^{ee} cannot be implemented, respectively. These simulations assume that $\delta^{ee} = 0$, $\beta(1+r) = 1$ and that productivity is constant.

Part 2 from proposition 1 considers what happens when some transfers can no longer be implemented. It states that the exact same allocations for consumption c_t and EE probability p_t as in the relaxed contract of part 1 can be achieved provided that assets are used to substitute for the missing transfer. The blue and orange lines in figure 1 show the implementation of the contract when the transfer τ_t^{eu}, or τ_t^{ee} cannot be implemented, respectively. The left and middle panels show that the paths of wages and assets differ across these implementations whereas the right panel confirms that the path of consumption remains the same.

Consider first the blue line, which shows how the optimal contract is implemented when $\tau_t^{eu} = 0$. Wages are *frontloaded* so that workers accumulate assets while employed. The firm thus helps workers self-insure against unemployment risk by making sure that

[8]The literature on optimal wage contracts assumes that workers cannot commit to transfers post EE transitions, τ_t^{ee}, not only because it is realistic but also because it is critical to generate wages that are backloaded, which is a key prediction of these models.

workers have enough assets to smooth consumption themselves if they become unemployed. If, instead, the transfer τ_t^{eu} could be implemented, firms would make a transfer to workers in the event of an exogenous separation into unemployment, effectively giving workers severance payment, so workers would not need to accumulate precautionary savings. Remarkably, the blue line provides a good description of the wage contracts observed in sectors where workers can commit to transfers post EE transitions. Barbu and Wang (2026) show that in specific sectors, such wage contracts are implemented by firms making loans to workers and these loans are enforced because the workers lose their license if they do not repay them when they switch jobs. In these sectors, the authors find that upfront hiring bonuses are prevalent, consistent with the blue line from figure 1. This evidence confirms that firms would frontload wages if they were not worried about worker retention, which is a critical force in my model.

Consider now the orange line, which shows how the optimal contract is implemented when $\tau_t^{ee} = 0$. Wages are *backloaded* and, in fact, negative during the first period of employment, and the worker borrows. To understand how assets are used here to influence worker mobility, it is useful to first consider how the optimal contract uses the transfer τ_t^{ee} when it can be implemented. The key contracting friction in the optimal contract is that the search decision of workers is private information. This means that the worker's search decision, and the implied EE probability, are chosen to maximize the worker's value, not the joint value from the match. Often, in the context of optimal wage contracts, this means that firms would like to reduce worker mobility. When the transfer τ_t^{ee} can be implemented, the optimal contract requires workers to make a payment back to their previous employer when they switch jobs. This transfer reduces the worker's continuation value at the next job and ensures that the worker's search decision maximizes the joint value from the current match. When the transfer τ_t^{ee} cannot be implemented, the optimal contract instead backloads wages to reduce the worker's assets as in figure 1. This too reduces the worker's continuation value at the next job and ensures that the worker's search decision maximizes the joint surplus.

Part 2 from proposition 1 requires productivity to be constant because transfers can be state-contingent whereas assets are chosen one period in advance, before productivity shocks are realized. It also requires a relaxed borrowing constraint because transfers can be negative, so implementing them using assets may require workers to borrow depending on their initial asset holdings. When $\delta^{ee} \neq 0$, a version of proposition 1 also holds but it requires introducing transfers after exogenous EE transitions.

Taken together these results show that assets are used to substitute for missing transfers post EE and EU transitions. In the model I estimate, no transfer can be implemented

and workers face borrowing constraints so assets cannot perfectly substitute for these missing transfers. As a result, firms face a trade-off when they choose how to use workers' assets in the optimal contract, which I characterize in the next section.

3.2 Implications for tenure profiles

This section uses the optimality conditions of the contract (3), including a new pseudo Euler equation, to characterize how assets influence wage contracts. In this section, I focus on tenure profiles that describe how wages, consumption and assets change over time when productivity remains constant. In the next section, I will discuss how firms pass productivity shocks through to workers.

The consumption growth condition Our starting point is

$$\frac{1}{u'(c_t)} - \frac{1}{u'(c_{t-1})} = -\frac{p_W(W_t, a_t)}{1 - p(W_t, a_t)} \left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}} \left[\Pi(s_{t+1}) | x_t \right]}{1+r} \right) + \frac{\Phi(W_t, a_t)}{u'(c_{t-1})} \tag{5}$$

where $\Phi(W_t, a_t) \equiv \beta(1+r) \left[1 + \frac{\delta^{ee}}{1-\delta^{ee}} \frac{1}{1-p(W_t, a_t)} \right] - 1$. This equation, derived in appendix A.2 from the contract (3), is similar to Burdett and Coles (2003), Shi (2009) and Balke and Lamadon (2022), except that consumption c_t replaces wages w_t on the left-hand side, and that the term $\Phi(W_t, a_t)$ appears on the right-hand side.

Equation (5) equates the benefits of backloading wages on the right to the costs on the left. Backloading wages means reducing w_{t-1} to increase w_t, and thus W_t. It has three benefits. The first, which is standard in wage contracts, is to reduce the worker EE transition rate. This depends on the extent to which promising workers a higher continuation value W_t reduces the EE rate, $p_W(W_t, a_t)$, and on the value that firms get from the match, $x_t - w_t + \mathbb{E}_{x_{t+1}} \left[\Pi(s_{t+1}) | x_t \right] / (1+r)$. The second benefit, in $\Phi(W_t, a_t)$, is to improve the worker's continuation value after exogenous EE transitions. In particular, an increase in continuation value W_t is valued by workers if they remain employed at the same firm but also by workers who switch jobs because it increases the continuation value at the next job W_t. By contrast, firms do not bear the cost of such promises after exogenous EE transitions because they are paid by poachers. This induces firms to backload wages more than they would without exogenous EE transitions ($\delta^{ee} = 0$). Finally, the third benefit of backloading wages, also in $\Phi(W_t, a_t)$, is to transfer value from workers to firms early on during the match when firms are less patient than workers ($\beta(1+r) > 1$), and inversely if workers are less patient than firms ($\beta(1+r) < 1$). In existing contracting models, firms and workers have the same discount factor and there are no exogenous EE transitions

so $\Phi(W_t, a_t) = 0$. In my quantitative model, it turns out that $\Phi(W_t, a_t)$ is almost exactly equal to 0 because the motive to backload wages due to exogenous EE transitions offsets the motive to frontload wages due to worker impatience almost perfectly.

Backloading wages is costly because it generates a gap in the marginal utility of consumption of workers over time. When workers are hand-to-mouth, backloading wages implies backloading consumption because $c_t = w_t$ so the cost of backloading is straightforward and equation (5) is enough to characterize the optimal contract, together with the definition of the worker and firm values $W_t, \Pi(s_t)$. By contrast, when workers can trade risk-free bonds, consumption and wages are not always equal so equation (5) must be combined with a new optimality condition relating wages, consumption and assets. This new condition is the pseudo Euler equation that I derive next.

The pseudo Euler equation This equation is derived from the optimality condition for assets in the optimal contract. The surprising result is that it takes the familiar form of an Euler equation, except for an additional term that arises because assets influence directly the job finding rate of workers. This result is not obvious from inspection of the optimal contract (3) because the first order condition includes terms like $S_a(W_{t+1}, a_{t+1})$, which describes how the worker surplus from EE separations depends on assets. This term in turn depends on the fact that assets influence the search decision of workers $v(W_{t+1}, a_{t+1})$ and the matching rate in each market $\lambda_w(v_{t+1}, a_{t+1})$. I now manipulate this optimality condition to show how it turns into the pseudo Euler equation from proposition 2, and then use this equation together with equation (5) to analyze how wages, consumption and assets change with tenure and over the life-cycle.

The first step to derive the pseudo Euler equation is to understand how assets influence job finding rates. For this, I turn to the free entry condition (4) that relates the value of new matches Π_0 and the vacancy filling rate λ_f (and thus the job finding rate λ_w through the matching function). First, notice that $\Pi_0(v, a) = \Pi_0(v, 0) + (1 + r)a$ because assets enter only the budget constraint of the worker in the initial period of the contract. This implies that the value of new matches increases in the asset of the worker. Next, notice that the firm value Π_0 is strictly decreasing in the worker's promised value v from the envelope condition. This suggests the existence of an indifference condition for firms between matching with a worker in a market with high value and high assets, and in a market with low value but also low assets. This also means that in equilibrium firms post relatively more vacancies in markets where workers ask for low values v, and in markets where they have high initial assets a. This relation is formally captured by the following

equation, derived by combining the free entry condition with the envelope conditions,

$$\partial_a \lambda_w(v_{t+1}, a_{t+1}) = -\partial_v \lambda_w(v_{t+1}, a_{t+1})(1+r)u'(c^{ee}_{t+1}) \tag{6}$$

where c^{ee}_{t+1} is the consumption of the worker at the next job after an EE separation. This equation states that, in equilibrium, increasing the assets of workers by Δa has the same effect on the EE probability as inducing workers to search in a market where the worker value is $(1+r)u'(c^{ee}_{t+1})\Delta a$ lower.

The next step is to combine this equation with the optimality condition for search v

$$\begin{aligned} S_a(W_{t+1}, a_{t+1}) &= \kappa \partial_a \lambda_w(v_{t+1}, a_{t+1}) \left[v_{t+1} - W_{t+1}\right] \\ &= -\kappa \partial_v \lambda_w(v_{t+1}, a_{t+1})(1+r)u'(c^{ee}_{t+1}) \left[v_{t+1} - W_{t+1}\right] \\ &= p(W_{t+1}, a_{t+1})(1+r)u'(c^{ee}_{t+1}) \end{aligned}$$

The first line uses the definition of $S(W, a)$ and the envelope theorem, the second line uses equation (6) and the third line uses the optimality condition for search v. Intuitively, raising the assets of workers increases their job finding rate so workers respond optimally by searching for jobs with a higher value, and thus a higher consumption at the next job. Thus, this equation shows that under the optimal contract increasing the worker's assets marginally by Δa has a similar effect on the worker value as increasing consumption at the next job by $(1+r)\Delta a$, which is what an Euler equation would imply. This is why the optimality condition for assets a_{t+1} in the optimal contract can ultimately be written as a pseudo Euler equation.

Proposition 2. *The optimal contract satisfies a pseudo Euler equation*

$$\begin{aligned} \tfrac{u'(c_t)}{\beta(1+r)} \geq \; & \delta u'(c^u_{t+1}) + (1-\delta)\delta^{ee}\mathbb{E}_{x_{t+1}}[u'(c_{t+1})|x_t] \\ & + (1-\delta)(1-\delta^{ee})\mathbb{E}_{x_{t+1}}\left[p(W_{t+1}, a_{t+1})u'(c^{ee}_{t+1})|x_t\right] \\ & + (1-\delta)(1-\delta^{ee})\mathbb{E}_{x_{t+1}}\left[(1-p(W_{t+1}, a_{t+1}))u'(c_{t+1})|x_t\right] + \mathcal{W}_t \end{aligned} \tag{7}$$

where $\mathcal{W}_t$ represents how assets influence the EE probability directly, and is equal to

$$\begin{aligned} \mathcal{W}_t \equiv - \; (1-\delta)(1-\delta^{ee})\tfrac{u'(c_t)}{\beta(1+r)}\mathbb{E}_{x_{t+1}}\Bigg[& \left(u'(c_{t+1})p_W(W_{t+1}, a_{t+1}) + \tfrac{p_a(W_{t+1}, a_{t+1})}{1+r}\right) \\ & \times \left(x_{t+1} - w_{t+1} + \tfrac{\mathbb{E}_{x_{t+2}}[\Pi(V_{t+2}(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\Bigg] \end{aligned}$$

Equation (7) holds with equality when the borrowing constraint (BC) does not bind.

Proof. See appendix A.3. □

Equation (7) contains two groups of terms. The first group includes everything but $\mathcal{W}_t$ and constitutes a standard Euler equation. It equalizes, up to $\beta(1+r)$, the worker's marginal utility of consumption today and tomorrow after exogenous EU transitions, after exogenous EE transitions, after endogenous EE transitions and within the same job. The second group of terms, gathered in $\mathcal{W}_t$, shows how assets influence the EE probability directly given a fixed path of wages. Quantitatively, virtually all the contract dynamics are driven by the first group of terms representing an Euler equation so I defer the discussion of $\mathcal{W}_t$ to section 5[9].

Implications for tenure profiles We are now ready to combine equations (5) and (7) to analyze how assets influence the tenure profiles for wages and consumption.

Equation (7) states that, given a path for wages, the optimal contract uses the worker's assets to smooth consumption over time within matches, but also across states after EU or EE separations. Consider for instance the standard case where wages are backloaded, meaning that they grow with tenure. Firms can replicate the allocation with hand-to-mouth workers and set $c_t = w_t$, but this would make consumption grow over time. As a result, contracts would be relatively unattractive to workers with concave utility who would prefer consumption to be stable over time. Firms would then have to pay workers higher average wages in order to attract them. Instead, firms could make the contract more attractive by doing what equation (7) prescribes. In particular, the firm could deplete the worker's existing assets to smooth consumption over time. Of course, firms would not want to deplete the worker's assets too much because workers also value precautionary savings against unemployment risk and because of borrowing constraints.

Consider now how firms set wages over time, knowing that they can smooth consumption using the worker's assets. The ability to use assets to smooth consumption reduces the cost of backloading wages on the left-hand side of equation (5). As a result, firms backload wages even more. If workers did not face any unemployment risk and could borrow, we would recover the special case of proposition 1 where wages are extremely backloaded and assets are used to substitute for transfers post EE transitions. In the presence of borrowing constraints, the worker's initial assets are critical because they influence firms' ability to smooth consumption, and thus to backload wages. If the worker has few initial assets, firms cannot backload wages too much because the borrowing constraint will bind. By contrast, if the worker has a lot of initial assets firms can backload wages more because the borrowing constraint is less likely to bind. Hence, a

[9] A sign that this additional term is irrelevant for contracts is that $\mathcal{W}_t / u'(c_t)$ equals 0.009% on average in simulations of the model with observable savings. Section 5 establishes this result formally by solving the model with hidden savings.

worker who has more assets to start with has access to a broad set of feasible contracts, and will thus search for different jobs than a worker with few assets.

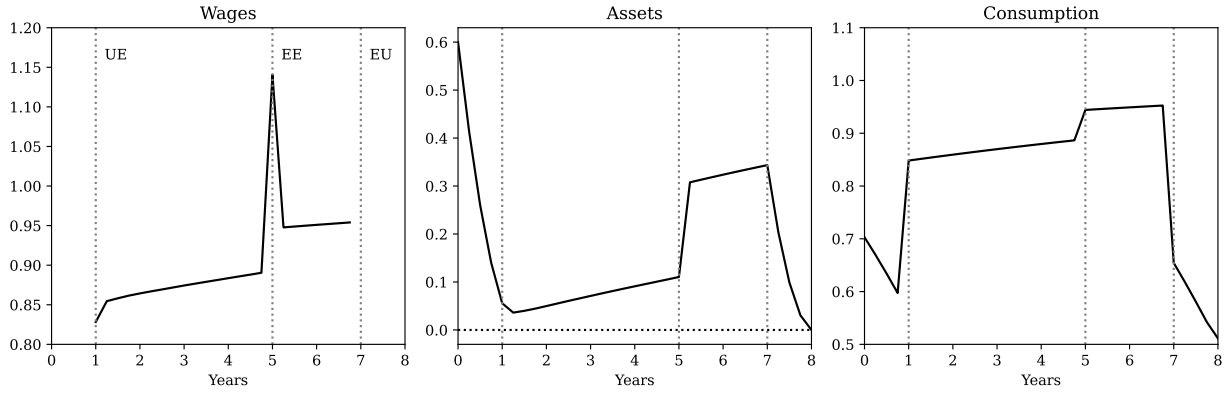
A novel implication of the model is that the worker's ability to self-insure against unemployment depends on the wage contract they receive from firms. In particular, firms' desire to retain workers makes them backload wages, which makes it difficult for workers to accumulate precautionary savings. When the firm's desire to retain workers is small, or when unemployment risk is large, wages are less backloaded and may even be frontloaded. In fact, if firms did not care about worker retention at all, we would recover the special case of proposition 1 where wages are extremely frontloaded and assets are used to substitute for transfers post EU transitions.

Life-cycle dynamics Figure 2 illustrates the implications of the optimal contract for the life-cycle dynamics of wages, assets and consumption in the quantitative model. In this specific example, the worker starts unemployed at $t = 0$ with some initial asset. The worker finds a job at year 1, makes an endogenous EE transition at year 5 and an EU transition at year 7. The left panel shows that wages increase during the first job between years 1 and 5. The worker then receives a hiring bonus when switching jobs, and wages mildly increase later on during the second job between years 5 and 7. The middle panel shows that the worker depletes assets when unemployed and accumulates assets when employed. Finally, the right panel shows that consumption falls while the worker is unemployed and rises while the worker is employed. Consumption jumps during UE, EE and EU transitions and is smooth otherwise.

What accounts for the life-cycle dynamics in figure 2, especially those between years 1 and 7 when the worker is employed? Given the path of wages, the paths of assets and consumption satisfy the pseudo Euler equation (7), meaning that workers smooth consumption over time and state. The path for wages reveals how firms balance their desire to retain workers with their desire to insure them. Specifically, wages are backloaded during the first job. The reason is that workers hired from unemployment start with a low promised value so the value that the firm gets from this match is high. Besides, the EE transition rate of these workers, who are cheap to poach, is high so backloading their wages leads to a large reduction in their quit rate. This implies that the benefits of backloading wages on the right of equation (5) are large. Furthermore, the income of these workers does not fall much if they become unemployed so they do not want to accumulate much precautionary savings yet. This means that the cost of backloading wages on the left of equation (5) is small. These observations imply that the net benefits of backloading wages are strong for workers hired from unemployment. Backloading wages

means that workers are initially hired with low wages but promised higher wages in the future, so the wage of workers increases over time and the worker starts accumulating precautionary savings. At year 5, the worker finds another job and receives a hiring bonus, which leads to the jump in consumption and precautionary savings. This hiring bonus effectively means that wages are frontloaded at the second job. The reason is that the precautionary savings motive is now much stronger than the retention motive so the new firm chooses to attract the worker by offering insurance against unemployment risk in the form of frontloaded wages.

Figure 2: Life-cycle dynamics for wages, assets and consumption



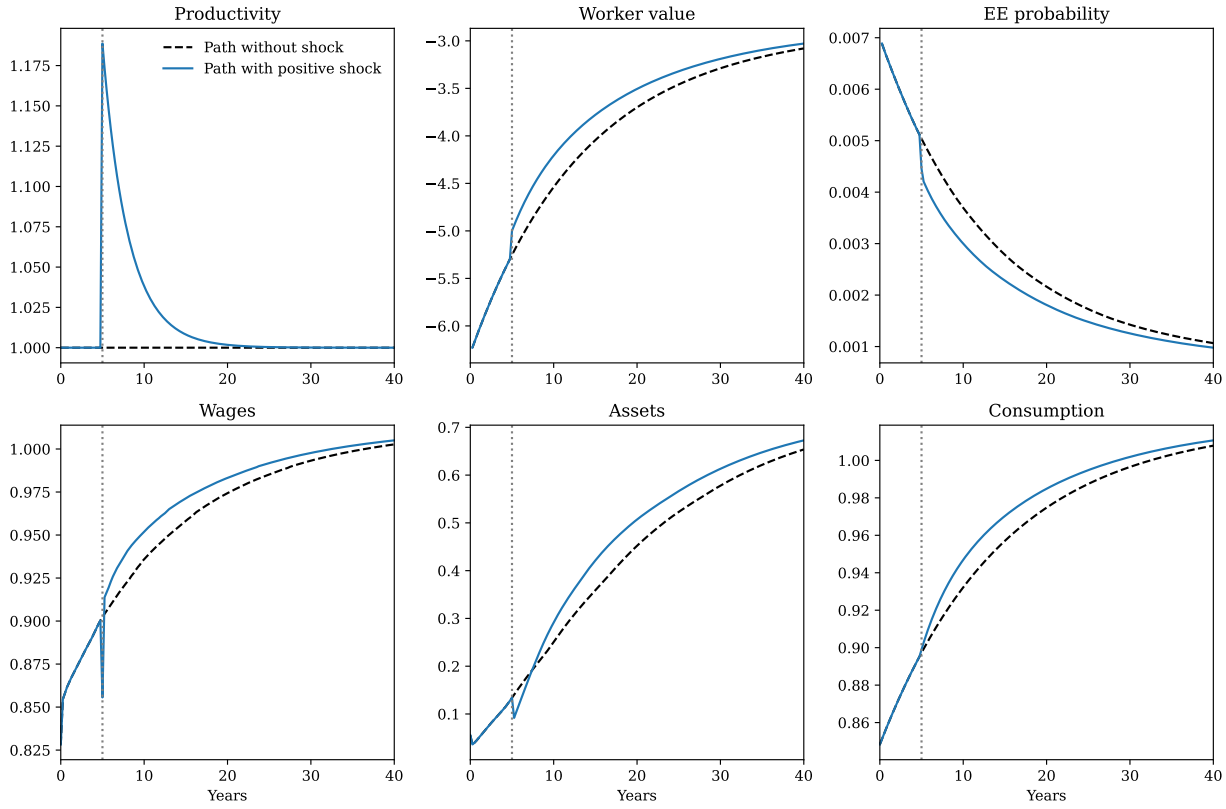
Note: example of paths for wages, assets and consumption for a worker who is unemployed with $a_0 = 0.6$ at $t = 0$. The dotted vertical lines show when the worker experiences UE, EE and EU transitions. The paths are computed in the quantitative model where shocks to productivity are expected to occur but where the realized values are null.

In conclusion, these results show that letting workers trade risk-free bonds leads to much richer life-cycle dynamics for wages, consumption and assets. I explore some of these implications with the quantitative model in section 4.

3.3 Implications for the pass-through of productivity shocks

Firm-level productivity shocks are an important source of risk for workers. This section describes how firms pass these shocks through to workers. The key innovation of the model is that firms can now pass shocks through differently to wages and to consumption. I illustrate this result here by comparing the paths of wages, assets and consumption for a worker who experiences a firm productivity shock to those for a worker who did not experience such a shock, whereas section 4 will quantify *how much* shocks are passed through to wages and consumption.

Figure 3: The pass-through of productivity shocks to workers



Note: I first compute the paths for a worker hired from unemployment at $t = 0$, who happens to remain employed with the same firm throughout and for whom the realization of productivity shocks ν_t is null. I then compute the same paths but for a worker who experiences a positive productivity shock after 5 years of employment. The pass-through is the difference between the paths with and without shock. The shock has persistence ρ and size $2 \times \sigma$, where ρ, σ are estimated in the data.

Figure 3 shows the tenure profiles of a worker hired from unemployment and who happens to remain employed with the same firm throughout. The top row shows the paths of productivity x_t, the continuation value of the worker at the current job W_t and the endogenous EE rate p_t while the bottom row shows the path of wages w_t, assets a_t and consumption c_t. The dashed black line shows the paths when the realization of productivity shocks is null whereas the solid blue line shows the paths for a worker who experiences a positive productivity shock of size $2 \times \sigma$ and persistence ρ after 5 years of employment. The pass-through is the difference between the paths with and without shock. The pass-through to negative shocks, not shown, is symmetric.

The top row shows that firms respond to a positive shock by increasing the worker continuation value in order to reduce the EE probability. The bottom row shows how this pass-through is implemented. The shock leads to a persistent increase in worker wages, together with a persistent increase in assets and consumption. Three observations stand

out. First, the change in the path of wages is small relative to the size of the productivity shock, which arises because the model is estimated to match the small pass-through measured in the data. Second, the shock increases the worker consumption both at the current job and after EE and EU transitions since the worker assets increase. This means that firm shocks now have more persistent effects on worker consumption than in models with hand-to-mouth workers. Third, the firm increases worker retention after the shock not only by paying workers more overall, but also by backloading wages more. This is because such backloaded wage increases raise worker retention not only today, but also in the future when productivity remains high since workers must stay with their current employer to actually get paid more. In fact, wages become so backloaded after such a persistent productivity shock that they fall on impact. Meanwhile, workers reduce their assets temporarily in order to increase their consumption smoothly over time in anticipation of higher future wages.

Taking stock, firms take advantage of workers' access to financial markets to pass shocks through to worker wages and consumption differently. In section 4, I will quantify how much these shocks are passed through to wages and consumption.

4 Quantifying the sources of insurance

I estimate the model to quantify how much insurance workers receive inside and outside firms. I find that workers receive significant insurance both inside and outside firms, and the insurance workers receive outside firms significantly crowds out the insurance that they receive inside firms. This has implications for inequality and policy. First, relatively wealthy workers experience higher wage growth and pass-through than poor workers, consistent with empirical evidence. These wealthy workers, however, receive more insurance overall and wages that are higher on average. Second, in response to more generous public insurance policies, workers receive wage contracts that pay more on average, but that are also more backloaded and potentially more exposed to productivity shocks.

4.1 Estimation

I estimate the model at quarterly frequency using administrative data on labor income, worker mobility, and firm productivity from France.

Data I use French matched employer-employee data that contains information on wages, worker mobility and firm productivity. The data covers 1/12th of the French labor force

and the near-universe of firms between 2005 and 2019. I focus on prime age workers (25-55 years old) who work in the private sector in for-profit firms with at least 3 employees, and who have a full-time job with a permanent contract. The final sample is an unbalanced panel with approximately 500,000 workers and 100,000 firms per year. Appendix B.1 provides details on the data.

Estimation I set some model parameters externally and estimate the remaining ones by moment matching.

The parameters set externally are the utility function, the real interest rate r, and the matching function. The utility function is CRRA with coefficient $\gamma = 2$, as is standard in the macroeconomic literature. The real interest rate is 1% quarterly. The matching function is Cobb-Douglas, $\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f) = B\left(\phi_u + \kappa\phi_e\right)^{\nu} \phi_f^{1-\nu}$, where ν is set to 0.5 following Petrongolo and Pissarides (2001) and B is normalized to 0.5.

The other parameters are inferred by matching moments from the data with moments from a panel of workers simulated from the model. The estimated parameters are the worker discount factor β, the vacancy posting cost k, the search efficiency on the job κ, the value of home production b, the exogenous separation rate into unemployment δ and into employment δ^{ee}, the persistence and volatility of productivity ρ, σ and the volatility of i.i.d. measurement errors for annual productivity σ^{meas}. These 9 parameters are estimated using 9 moments in the data.

Table 1 shows the moments used in the estimation and the parameter values. From the administrative data, I measure the average duration of unemployment and the transition rates from employment (EE and EU rates), the wage tenure profile defined as the cumulative wage growth after 20 years of tenure estimated with controls for worker experience in the labor market, the wage pass-through defined as in Guiso et al. (2005) and the variance and autocorrelation of productivity growth measured as value added per worker. The annual marginal propensity to consume is set to match estimates from Blundell et al. (2008). Appendix B.2 provides details on these moments.

The combination of parameters and moments in the estimation is fairly standard for the literature on optimal wage contracts with search, with two exceptions. The first is to target a marginal propensity to consume that is lower than 1, which is possible only when workers can trade assets. The target of 40% annually is achieved by making workers relatively impatient $\beta(1+r) < 1$, which influences their demand for savings and the average marginal propensity to consume in equilibrium. This approach is standard in the precautionary savings literature (e.g. Auclert, Rognlie and Straub, 2024). The second is to introduce exogenous EE transitions with probability δ^{ee}. These transitions help account

Moments	Data	Model	Parameters	
Average unemp. duration, in quarters	5.2	5.2	Cost of posting vacancy k	3.33
Annual EE rate	7.1%	7.1%	Exogenous EE transition rate δ^{ee}	0.017
Annual EU rate	6.5%	6.5%	Separation rate δ	0.018
Wage tenure profile	10.3%	10.3%	Flow unemployment value b	0.51
Wage pass-through	5.8%	5.8%	On-the-job search efficiency κ	0.07
Marginal propensity to consume	40.0%	40.0%	Relative discount factor $\beta(1+r)$	0.983
Variance of productivity growth	0.041	0.041	Volatility of productivity σ	0.0864
Autocorrelation of order 1	- 0.25	- 0.25	Persistence of productivity ρ	0.926
Autocorrelation of order 2	- 0.067	- 0.067	Volatility of meas. error σ^{meas}	0.107

Note: the left panel shows the moments used in the estimation in the data and in the model and the right panel shows the parameters estimated internally. See appendix B.2 for details.

Table 1: Targeted moments in data vs. model and estimated parameters

for the small wage pass-through of 5.8% together with the annual EE rate of 7.1%. To see why, note that the small pass-through in the data suggests that workers receive significant insurance inside the firm. But in this model where workers also receive significant insurance from their assets, firms would rather optimize worker retention with a much higher pass-through. Introducing exogenous EE transitions with rate δ^{ee} makes worker mobility less responsive to wage contracts, which reduces the optimal degree of pass-through in the model. To jointly account for the pass-through and the EE transition rate, I find that exogenous transitions must account for the majority of EE transitions (5.9% out of 7.1%). Exogenous EE transitions are a common feature of dynamic search models (e.g. Jolivet et al. 2006) so it is not surprising that they also play a role here. Appendix B.3 provides an alternative specification in which all EE transitions are endogenous ($\delta^{ee} = 0$), and where the pass-through is small because of persistent measurement errors. All the results derived in this section are robust to this alternative specification.

Overall, in the estimated model workers receive substantial insurance both inside and outside firms. In particular, the small tenure profile and pass-through mean that firms provide substantial insurance to workers while the relatively low marginal propensity to consume means that workers receive significant insurance through their assets. The main validation of the model is to account for heterogeneity in wage growth and pass-through across the wealth distribution, which I describe in section 4.2.

Implications for contracts Table 2 compares key moments on wage contracts from my quantitative model and from a model with hand-to-mouth (HtM) workers. In the model with HtM workers, all parameters are kept constant except that $a_{t+1} = 0$ for employed

	Growth of w	Growth of c	Pass-through to w	Pass-through to c
Baseline	0.7%	0.6%	5.8%	2.3%
Model with HtM workers	0.6%	0.6%	2.5%	2.5%

Note: the top row reports moments computed in the baseline model and the bottom row reports the same moments in a model with hand-to-mouth workers. The model with HtM workers is calibrated using exactly the same parameters as the baseline model, except that the constraint $a_{t+1} = 0$ is imposed for unemployed and employed workers. The wage and consumption growth are calculated annually for continuously employed workers. The pass-through to wages and consumption is computed as in table 1.

Table 2: Comparison with HtM model

and unemployed workers. Wage and consumption growth are measured as annual growth rates for continuously employed workers. The pass-through of productivity shocks to wages and consumption is measured as in table 1.

The first column of table 2 shows that wages grow on average by 0.7% per year in the baseline model and by 0.6% per year in the model with HtM workers and the second column shows that consumption growth is identical across models (0.6%). Thus, introducing assets does not influence much the average tenure profile for wages but this masks substantial heterogeneity across workers, as I document in section 4.2. The third column from table 2 compares the pass-through of productivity shocks to wages across models. The estimated wage pass-through is 5.8% in the baseline model, more than twice the 2.5% estimated in the model with HtM workers. This means that workers receive much less insurance from firms when they can trade assets than when they are HtM. However, despite receiving less insurance from firms, workers receive more insurance overall in the baseline model. This can be seen from column 4, which shows that the pass-through to consumption is 2.3% in the baseline model, and 2.5% in the model with HtM workers. This is the result of differences in the marginal propensity to consume across models: it is equal to 40% when workers trade assets, and 100% when they are HtM.

These results show that the insurance that workers receive outside firms crowds out the insurance that they receive inside firms. The crowding out against shocks appears substantial in this exercise. To see this, consider the following back-of-the-envelope calculation: if letting workers trade risk-free bonds did not influence wage contracts, the pass-through would have remained equal to 2.5% in the baseline model; the pass-through to consumption would therefore have been equal to $2.5\% \times \text{MPC} = 2.5\% \times 40\% = 1\%$, which is less than half the actual pass-through in the baseline model (2.3%). One interpretation of this result is that workers are effectively less risk averse against shocks when they can trade assets, so firms choose to provide less insurance to them.

This crowding out appears when we compare the models with and without assets. In the next sections, I show that the same crowding out is at play within the baseline model

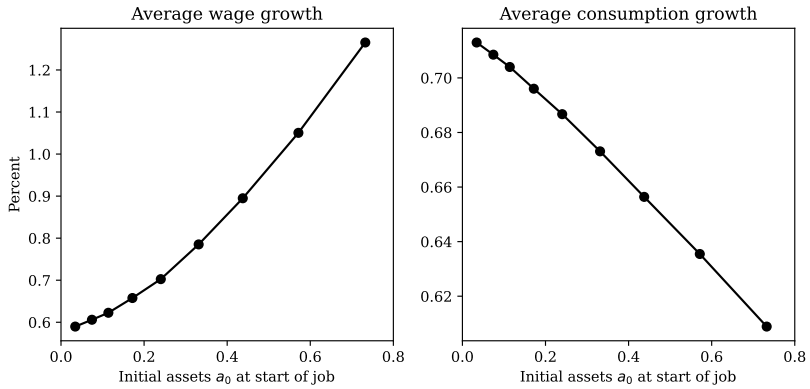
across workers with different assets and in response to policy changes.

4.2 Heterogeneity across the wealth distribution

I now show that workers across the wealth distribution receive different amounts of insurance inside and outside firms. This provides an explanation for the existing empirical evidence that wealthy workers experience higher wage growth and pass-through than workers with few assets.

Wage growth Figure 4 shows the average wage growth and consumption growth of workers who have different levels of assets when they get hired from unemployment. To compute this figure, I simulate a panel of workers starting unemployed with assets corresponding to different deciles of the stationary distribution. I then track these workers until the first separation into unemployment.

Figure 4: Heterogeneous earnings and consumption growth over the wealth distribution



Note: the statistics are computed by simulating a panel of workers starting unemployed at $t = 0$ with assets from different deciles of the stationary distribution. These workers are tracked until the first separation into unemployment. The left panel shows the average annual growth rate of wages and the right panel shows the average annual growth rate of consumption, computed as in table 2.

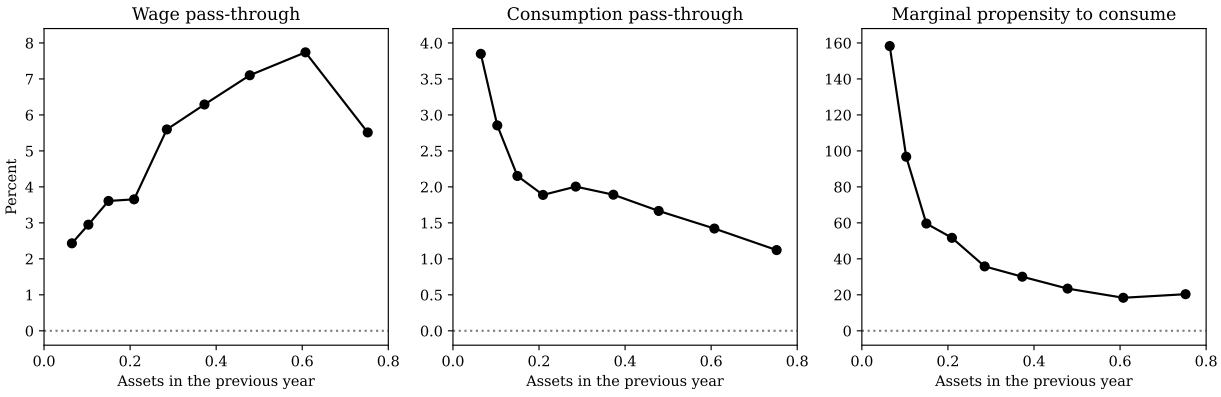
The left panel shows that wealthy workers, who start their employment with more assets, experience higher wage growth relative to poor workers who start their career with fewer assets. The differences in wage growth are substantial: workers with initial assets equal to about 2 months of average wage (top decile) experience an average wage growth that is about twice as large as that of workers with no asset (bottom decile). These differences arise because wealthy workers can use their initial assets to smooth their consumption over time, unlike poor workers who cannot. Firms therefore backload wealthy workers' wages more, providing them with less insurance, while offering poorer workers

more stable wages. Thus, the insurance that workers receive outside firms significantly crowds out the insurance they receive inside firms. The right panel shows that the consumption of wealthy workers is more stable over time than the consumption of poor workers. Thus, despite receiving less insurance inside firms, wealthy workers receive more insurance overall than poor workers.

This mechanism provides an explanation for existing evidence on earnings growth across the asset distribution. In particular, Halvorsen et al. (2022) show using Norwegian data that the children of parents with high net wealth experience higher wage growth than children of parents with low net wealth. Guiso et al. (2013) find using Italian data that firms experiencing liquidity constraints backload the wages of workers who are more likely to be wealthy, namely managers and white-collar workers relative to blue-collar workers. Critically, in these papers the prevailing wealth of workers (or their parents) predicts future income profiles, suggesting that wealth inequality leads to inequality in earnings. The same is true in my model: the wage contracts that workers receive depend on their assets when they start a job.

Pass-through of productivity shocks Figure 5 shows the pass-through of firm-level productivity shocks to wages and consumption and the MPC for workers with different levels of assets. To compute this figure, I consider the stationary distribution and rank workers according to their average assets in the previous year.

Figure 5: Heterogeneous pass-through over the wealth distribution



Note: the statistics are computed using the stationary distribution for each bin of the asset distribution in the previous year. The pass-through to wages and consumption and the marginal propensity to consume are computed as in table 2.

The left panel shows that the pass-through to wages generally increases with the worker assets, which means that wealthy workers receive less insurance inside firms. The middle panel shows that these workers nevertheless experience lower pass-through

to their consumption, which means that they receive more insurance overall. The right panel shows why: wealthy workers have a much lower MPC. Here again, these differences arise because firms pass shocks through more to the wages of wealthy workers who have more assets to absorb them. This crowding out is substantial: the pass-through to wages is twice as large at the top of the wealth distribution as at the bottom.

This mechanism provides an explanation for existing evidence on pass-through and MPC across the wealth distribution. In particular, Fagereng et al. (2018) show using Norwegian data that the pass-through of firm-level productivity shocks to wages is increasing in the prevailing financial assets of workers. In fact, in their data the pass-through is about twice as large at the top of the distribution as at the bottom, which is consistent with my results. The fact that the MPC is decreasing with assets is also consistent with a broad set of evidence (e.g. Blundell et al., 2008).

Compensation for risk A surprising implication of the model is that wealthier workers also receive higher wages on average, precisely because they receive less insurance from firms. To establish this result, I first simulate the employment history of workers who get hired from unemployment and who are initially in the bottom decile of the asset distribution with almost no initial assets ($a_0 = 0.03$). I track these workers until their first separation into unemployment. I repeat this simulation for workers initially in the top decile who have approximately 2 months of earnings as assets ($a_0 = 0.73$). Finally, I compare the average earnings of wealthy workers to those of poor workers. In this exercise, I restrict wealthy workers to initially search for jobs in markets with the same job finding rate as workers with few assets to ensure that my results are not driven by wealth effects on search and instead by the wage contracts received by these different workers.

I find that relatively wealthy workers get paid on average 0.85% more than poor workers. Thus, wealth inequality generates income inequality because wealthy workers receive different wage contracts than poor workers. This difference in average wages arises because firms expect higher profits from matching with wealthy workers and, in equilibrium, these higher expected profits translate into higher wages because of the free entry condition. There are two reasons why firms expect higher profits from wealthy workers. The first is that their profits are more frontloaded, because the wages they offer to workers are more backloaded. The second is that they are able to retain workers more, especially when productivity is high. These differences in average wages are quantitatively modest, but they are nonetheless essential because they induce wealthy workers to search for contracts with less insurance. Said differently, these differences are a compensation for risk

	Growth w	Growth c	Pass-through w	Pass-through c	Average w
Baseline	0.7%	0.6%	5.7%	2.3%	-
Relaxed borrowing ($\underline{a} = -1$)	2.3%	0.5%	6.3%	2.2%	+1.08%
Higher benefits (+50%)	0.9%	0.6%	3.3%	2.5%	+0.19%

Note: the first row considers the baseline model without borrowing and with unemployment benefits b. The second row considers the model in which workers can borrow up to $\underline{a} = -1$, which corresponds to approximately 1 quarter of average labor income. The third row considers the model in which unemployment benefits are higher by 50% for one year on average. The statistics are computed for simulations where workers are hired from unemployment and are followed until the first separation into unemployment. The first two columns report the average growth rate of wages and consumption. The next two columns report the pass-through of wages and consumption, computed as in table 2. The last column reports the change in average wage for these workers.

Table 3: Public insurance policies

that wealthy workers receive from firms[10].

To conclude, these results show that the insurance that workers receive outside firms crowds out the insurance that they receive inside firms. This provides an explanation why wealthier workers experience higher wage growth and high pass-through in the data. I find that wealthier workers not only receive more insurance overall but also higher wages on average as a compensation for the lower insurance they get from firms.

4.3 Public insurance policies

In this section, I show that public insurance policies that improve how much insurance workers receive outside firms also crowd out the insurance that workers receive inside firms. Specifically, I compare the effects of expanding household access to credit and of increasing unemployment benefits.

Expansion in household access to credit I model an expansion in household access to credit by lowering the borrowing constraint from $\underline{a} = 0$ to $\underline{a} = -1$, which is about 1 quarter of wages. This policy could be the result of a subsidy or public guarantee on borrowing aimed at helping workers smooth their consumption over the life-cycle or against shocks. In this counterfactual, I assume that workers can borrow at rate r and all other parameters are kept constant. I simulate a panel of workers starting from unemployment with initial assets drawn from the stationary distribution and track these workers until the first separation into unemployment.

[10]This effect is quantitatively comparable to wealth effects on search described in Eeckhout and Sepahsalari (2024) and Chaumont and Shi (2022) but it operates through a different mechanism. In these papers, search is directed and workers receive fixed-wage contracts. Wealthy workers receive higher wages because wealth effects make them search in markets with a lower job finding rate and higher wages. By contrast, in my model wealthier workers receive higher average wages even when the job finding rate is kept constant.

The second line of table 3 presents the result of this exercise. Relative to the baseline, workers are only slightly better insured overall when borrowing constraints are relaxed. The reason is that firms respond to the policy by providing much less insurance to workers. To see this, note that the growth rate of consumption is slightly lower when workers can borrow (0.5% compared to 0.6%) whereas the growth of wages is much higher (2.3% compared to 0.7%). Similarly, the pass-through of productivity shocks to consumption is slightly lower (2.2% compared to 2.3%) whereas the pass-through to wages is higher (6.3% compared to 5.7%[11]). Thus, the improved insurance that workers receive outside firms, by borrowing, significantly crowds out the insurance that they receive inside firms, from wage contracts. In fact, the crowding out offsets most of the effects of the policy in terms of smoothing consumption. In addition, the last column of table 3 shows that workers now receive wages that are 1.08% higher on average when they can borrow. As in section 4.2, this arises because firms provide less insurance to workers. Thus, workers gaining access to borrowing should expect to receive wages that are more backloaded and more volatile, but also slightly higher on average.

Higher unemployment benefits It is well known that unemployment benefits influence the asset accumulation of workers who self-insure against unemployment risk. In particular, when benefits are low workers have to postpone consumption in order to accumulate precautionary savings, especially when they are young and have few assets. An increase in unemployment benefits can thus enable workers to smooth their consumption over time instead of saving their way out of the borrowing constraint. In this exercise, I evaluate whether these conclusions are robust to the introduction of optimal wage contracts. That is, I quantify how a change in unemployment benefits influences the consumption of employed workers before they become unemployed.

I consider an increase in unemployment benefits to $1.5b$ for newly separated workers. Starting at separation, the increase expires with probability $1/4$ each quarter, after which benefits return to b. Workers therefore receive higher benefits for one year on average. I track the employment history of workers who are initially unemployed with standard benefits b and initial assets drawn from the stationary distribution until the first separation into unemployment. Keeping initial benefits at b limits the change in workers' initial search decisions, in terms of value and job finding rate. The higher benefits, however, influence how firms choose to deliver a given value to workers, and in particular how backloaded wages are. Studying instead a permanent increase in benefits b would make

[11] Table 3 tracks workers until the first separation into unemployment, so the sample is slightly different than the stationary distribution used in table 2. This explains the small differences in pass-through.

it harder to interpret the results because it would also influence how selective workers initially are when they search for a job from unemployment.

The third line of table 3 presents the result of this exercise. The growth rate of wages increases (0.9% compared to 0.7%) whereas the growth rate of consumption remains the same (0.6%). This arises because workers want to accumulate less precautionary savings when unemployment benefits are higher, so firms respond by backloading wages more. The effect of unemployment benefits on wage growth is smaller than the effect of relaxed borrowing because most workers are already at their borrowing constraint when they start a job, so firms cannot backload wages much more than they already are. Consider now the effect on the pass-through. The pass-through to wages is significantly lower in the counterfactual economy (3.3% compared to 5.7%). The reason is that, as workers accumulate fewer assets against unemployment risk, they are also less insured against productivity shocks. Firms thus choose to provide more insurance to workers by lowering the pass-through to wages. Overall, the changes in the consumption growth rate and pass-through are minimal, which shows that endogenous contracts offset most of the effects of the policy on employed workers. Finally, workers receive higher wages on average (+0.19%) but the effects are smaller than for borrowing constraints because firms still provide significant insurance to workers.

In conclusion, public insurance policies that improve how much insurance workers receive outside firms crowd out the insurance that they receive inside firms. These policies, however, can have opposite effects on the different sources of risk that workers face (unemployment risk, productivity shocks). Evaluating these policies in a model where wage contracts are exogenous would not only overstate their benefits in terms of insurance, but also understate their benefits in terms of average wages.

5 Hidden assets

So far I have assumed that the initial assets of newly hired workers a_0 and their savings decision a_{t+1} were public information to firms. These assumptions were critical to keep the problem tractable. Without them, we would need to consider that workers might misreport their assets when they meet a new firm and make different savings decisions than the ones recommended by the contract. A natural concern is whether my results are robust to relaxing these assumptions. In this section, I show that my results are, in fact, robust. I reach this conclusion first by relaxing the assumption that the savings decision is public information and then by also relaxing the assumption that the worker's initial assets are public information. I focus the discussion here on the results and their

interpretation, and leave technical details to appendix C.

5.1 Hidden savings a_{t+1}

Hidden savings lead to deviations by workers as a result of standard wealth effects on search and of a new effect of assets on wage backloading. Despite these forces, I find that the wage dynamics from the model with hidden savings are quantitatively indistinguishable from those from the model considered so far with observable savings.

Deviations due to hidden savings arise when the worker's preferred savings decision differs from that recommended in the optimal contract. This can be seen from equation (7) where the term $\mathcal{W}$ creates a wedge between the contract optimality condition with observable savings and an Euler equation that satisfies the worker's privately optimal savings decision. When $\mathcal{W} = 0$, the optimal contract satisfies the worker's optimality condition so the equilibria with observable savings and private savings coincide[12]. To understand the term $\mathcal{W}$, consider a firm that is contemplating a marginal increase in the savings of workers a_{t+1}, keeping the path of wages constant. This induces an increase in the continuation value of workers at the current job W_{t+1} by $(1+r)u'(c_{t+1})$, and therefore a reduction in the EE probability of $(1+r)u'(c_{t+1})p_W(W_{t+1}, a_{t+1})$. At the same time, this increase in assets makes it easier for workers to find a new job, which increases the EE probability by $p_a(W_{t+1}, a_{t+1})$. When these two effects exactly offset, $\mathcal{W} = 0$ and a change in savings has no direct effect on the worker EE probability *when the path of wages is held constant*. In this case, firms cannot influence the EE probability directly by manipulating assets so their recommendation for savings coincides with the worker's optimality condition. Introducing assets would then influence contracts only to the extent that it induces firms to change the path of wages, as explained in section 3.

In my model, deviations due to hidden savings arise for two reasons. The first is that the worker's assets have wealth effects on search (Acemoglu and Shimer, 1999), which is well known in the optimal unemployment insurance literature. In this literature, the equilibrium with hidden savings is exactly equivalent to the equilibrium with observable savings when the utility function is CARA, $u(c) = -\exp(-\gamma c)/\gamma$, because this utility function removes those wealth effects (Shimer and Werning, 2008). By contrast, in my model this equivalence breaks down even with CARA utility because the assets of workers also influence how backloaded wages can be after an EE transition. In particular, wealthy workers have access to a broader set of contracts than poor workers and thus face a different trade-off between higher job finding rates and higher wages. This is the

[12] This assumes that the first-order approach holds, which is not always true (Kocherlakota, 2004).

second, and new, reason why hidden savings lead to deviations in my model.

To assess the quantitative importance of these deviations, I solve the dynamic model under the assumption that the savings decision of workers is private information using the first-order approach[13] (Werning, 2001, Ábrahám and Pavoni, 2008). This approach consists in solving the optimal contract (3) with the Euler equation as an additional constraint and the worker's promised marginal utility of consumption as an additional state variable. I find that the paths of wages, consumption and worker mobility in this model are virtually indistinguishable from those in my baseline model where the savings decision is public information. For example, the path of wages differs by at most 0.03% for a worker hired from unemployment.

To summarize, the model with observable savings considered in this paper is a very good approximation to a model with hidden savings.

5.2 Hidden initial assets a_0

I now revisit the assumption that firms can observe the initial assets of newly matched workers. In practice, firms can approximate these assets with the information available to them. For example, firms know about the education and occupation of workers, both of which are correlated with wealth. Besides, firms typically know how long a worker has been unemployed before a match. Firms can thus design wage contracts given this information, for example by offering relatively more backloaded wage contracts to workers in wealthier occupations. However, even within occupations there is still significant heterogeneity in wealth across workers so it is natural to question whether the equilibrium changes when the worker's initial assets a_0 are private information.

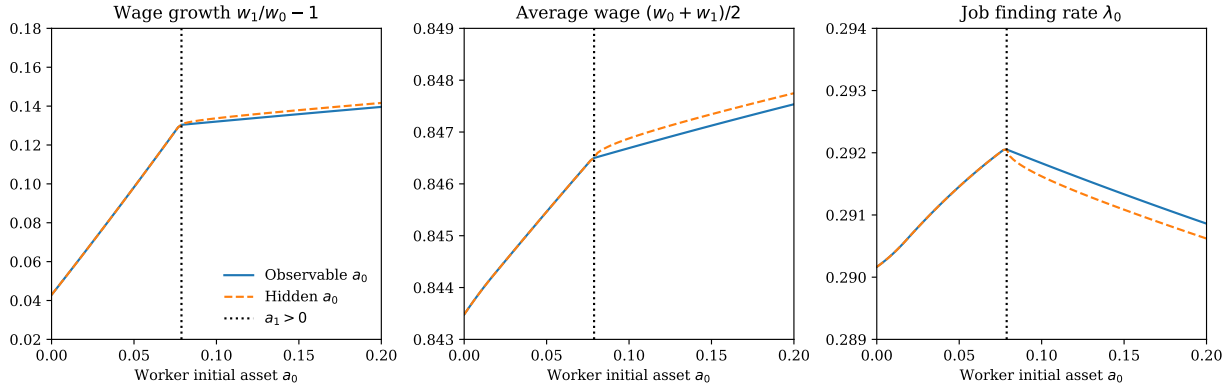
To answer this question, I compute the equilibrium in which assets are private information using the methodology of Guerrieri et al. (2010). This is now an adverse selection problem, where the worker's hidden type is the initial asset, with moral hazard, where both the savings and search decisions of employed workers are private information. To keep this problem tractable, I consider a 2-period version of my model with constant productivity[14], which I describe in appendix C.2. I establish two results. First, the novel implication that workers with different initial assets a_0 experience different wage growth does not hinge on the assumption that firms observe assets. Second, a small modification

[13]This model has to be solved with high numerical accuracy because the difference with the baseline model assuming observable savings is extremely small. For this reason, I focus on a simpler version of the model without productivity shocks and keeping the general equilibrium job finding rate $\lambda_w(v, a)$ and unemployment value $U(a)$ from the model with observable savings. See appendix C.1 for details.

[14]The 2-period model can be solved numerically without recursive methods. Solving the dynamic model with hidden search, hidden savings and hidden initial assets is very challenging.

to optimal contracts induces workers to reveal their initial assets truthfully.

Figure 6: Optimal contract with hidden initial assets a_0



Note: the figure compares the equilibrium in which the worker's initial asset a_0 is public information and the separating equilibrium in which it is private information but workers reveal their assets truthfully to firms. This exercise is done in a 2-period model with constant productivity where unemployed workers can find a job in period 0 and switch jobs in period 1. In both versions, the savings decision of workers is private information. The left panel reports the growth rate of wages of workers who find a job at $t = 0$, the middle panel reports their average contractual wage, and the right panel the job finding rate from unemployment at $t = 0$. The calibration is CRRA utility with $\gamma = 2$, $r = 0$, $\beta = 1$, $\delta = 0.1$, $b = 0.7$, $k = 0.25$, and $\kappa = 1$. These results show that the equilibrium does not unravel if initial assets are private information.

Figure 6 compares the optimal contracts that workers with different initial assets a_0 receive under the assumption that their initial assets are observable (solid blue line) or private information (dashed orange line). To the left of the vertical line, the borrowing constraint binds and workers do not save. In that region, workers with more assets receive wages that are more backloaded so they experience higher wage growth (left panel). As a result, these workers enjoy wages that are higher on average (middle panel) and a higher job finding rate from unemployment in the first period (right panel). To the right of the vertical line, the savings of employed workers are positive. In that region, the worker's initial assets no longer influence how backloaded wages can be because the borrowing constraint does not bind. They still influence contracts slightly because of standard wealth effects (e.g. Eeckhout and Sepahsalari, 2024): wealthy workers search for jobs with wages that are slightly more backloaded (left panel), with lower job finding rates (right panel) but that pay more on average (middle panel).

To understand the results in figure 6, it is useful to discuss first why workers would benefit from misreporting their initial assets. In this 2-period model, profitable deviations arise because of wealth effects on search with CRRA preferences, as in Chaumont and Shi (2022)[15]. Because of these wealth effects, workers with more assets during EE transi-

[15]They study a model with assets and directed search, but with constant wage contracts. In their model,

tions search in markets with higher values and lower job finding rates than poor workers. Firms expect workers with high initial assets to save more, and thus have more assets during EE transitions and be less likely to quit. In equilibrium, they are willing to offer these workers higher average wages. A newly matched worker thus benefits from overstating his initial assets slightly in order to receive higher wages on average. This worker would then save less and have a higher EE probability than expected by the firm[16].

Keeping this in mind, consider the region to the left of the vertical line in figure 6, where the equilibria with observable or hidden assets coincide. In that region, the savings of workers are already at the borrowing constraint so there are no gains from deviating and misreporting initial assets. Yet, in that region, workers receive wages that are more backloaded when they have more initial assets. This shows that this result has nothing to do with the observability of assets. It arises because workers across the wealth distribution voluntarily search for different contracts: wealthy workers can smooth their consumption better so they search for contracts with wages that are more backloaded because these jobs pay more on average and have a higher job finding rate.

Consider now the region to the right of the vertical line, where the equilibrium with hidden assets differs from the equilibrium with observable assets. In this region, firms induce workers to reveal their assets truthfully by backloading wages slightly more for wealthier workers (left panel) and by reducing their job finding rate (right panel). These changes induce firms to make higher expected profits so in equilibrium workers receive higher average wages (middle panel). The single crossing property that leads to separation is that reducing the job finding rate and making wage contracts more backloaded is especially costly to workers with few assets a_0. This prevents workers from pretending to be wealthier than they truly are. A key difference between my model and existing models with adverse selection and directed search is that separation is not only induced by changes in the job finding rate, but also by changes in the riskiness of contracts.

These results have important implications for the interpretation of the model. One valid interpretation is that workers report their assets to employers once they match, and then receive wage contracts tailored to their type. An equally valid, and more intuitive, interpretation is that different firms offer wage contracts with different degrees of insurance and that workers choose where to apply depending on their initial assets. Firms

workers also benefit from misreporting their initial assets to firms. They do not study the equilibrium with private information and instead argue that firms have enough information from worker employment history to back out what these assets are.

[16] In the 2-period model, the additional effect of assets on how backloaded wages can be at the next job is not present. This is because the worker can only switch jobs in the last period so the worker only receives a wage for one period after an EE transition.

never *ask* workers about their assets, but they have this information because they know which workers the contracts they offer will attract.

In conclusion, this section confirms that my baseline model with observable assets leads to similar answers to a model with private assets, which is arguably more realistic but also much more challenging to solve.

6 Conclusion

This paper builds a new equilibrium search model with optimal wage contracts and assets. This model brings together two perspectives on how workers get insured against labor market risk: inside the firm, through optimal wage contracts, and outside the firm, through assets and precautionary savings. The insurance that workers receive outside the firm crowds out the insurance that they receive inside the firm. This model can be used as a foundation for future work studying the sources and consequences of fluctuations in earnings and consumption, for example over the business cycle.

References

Ábrahám, Árpád and Nicola Pavoni, "Efficient allocations with moral hazard and hidden borrowing and lending: A recursive formulation," *Review of Economic Dynamics*, 2008, *11* (4), 781–803.

_ and Sarolta Laczó, "Efficient Risk Sharing with Limited Commitment and Storage," *The Review of Economic Studies*, July 2018, *85* (3), 1389–1424.

_ , Fernando Álvarez-Parra, and Susanne Forstner, "The Effects of Moral Hazard on Wage Inequality in a Frictional Labor Market," 2013.

Acabbi, Edoardo Maria, Andrea Alati, and Luca Mazzone, "The Collapse of Human Capital Ladders in Recessions," 2026.

Acemoglu, Daron and Robert Shimer, "Efficient Unemployment Insurance," *Journal of Political Economy*, October 1999, *107* (5), 893–928.

Agostinelli, Francesco, Domenico Ferraro, Giuseppe Sorrenti, and Leonard Treuren, "Employment Relationships, Wage Setting, and Labor Market Power," 2025.

Aiyagari, S. Rao, "Uninsured Idiosyncratic Risk and Aggregate Saving," *Quarterly Journal of Economics*, August 1994, *109* (3), 659–684.

Alves, Felipe, "The Job Ladder in HANK: Inspecting the Mechanism," 2025.

Andrews, Isaiah, Matthew Gentzkow, and Jesse M. Shapiro, "Measuring the Sensitivity of Parameter Estimates to Estimation Moments," *The Quarterly Journal of Economics*, November 2017, *132* (4), 1553–1592.

Auclert, Adrien, Matthew Rognlie, and Ludwig Straub, "The Intertemporal Keynesian Cross," *Journal of Political Economy*, December 2024, *132* (12), 4068–4121.

Azariadis, Costas, "Implicit Contracts and Underemployment Equilibria," *Journal of Political Economy*, December 1975, *83* (6), 1183–1202.

Azzalini, Gualtiero, "Business cycle asymmetry of earnings pass-through," 2023.

Baily, Martin Neil, "Wages and Employment under Uncertain Demand," *Review of Economic Studies*, January 1974, *41* (1), 37–50.

Balke, Neele and Thibaut Lamadon, "Productivity Shocks, Long-Term Contracts, and Earnings Dynamics," *American Economic Review*, July 2022, *112* (7), 2139–77.

Barbu, Alexandru and Yiran Wang, "Employee Forgivable Loans," 2026.

Bewley, Truman, "The Permanent Income Hypothesis: A Theoretical Formulation," *Journal of Economic Theory*, December 1977, *16* (2), 252–292.

Blundell, Richard, Luigi Pistaferri, and Ian Preston, "Consumption Inequality and Partial Insurance," *American Economic Review*, December 2008, *98* (5), 1887–1921.

Burdett, Ken and Melvyn Coles, "Equilibrium Wage-Tenure Contracts," *Econometrica*, 2003, *71* (5), 1377–1404.

Caratelli, Daniele, "Labor Market Recoveries Across the Wealth Distribution," 2025.

Chaumont, Gaston and Shouyong Shi, "Wealth Accumulation, On-the-Job Search and Inequality," *Journal of Monetary Economics*, 2022.

Cole, Harold L. and Narayana R. Kocherlakota, "Efficient Allocations with Hidden Income and Hidden Storage," *Review of Economic Studies*, July 2001, *68* (3), 523–542.

Eeckhout, Jan and Alireza Sepahsalari, "The Effect of Wealth on Worker Productivity," *The Review of Economic Studies*, 05 2024, *91* (3), 1584–1633.

Fagereng, Andreas, Luigi Guiso, and Luigi Pistaferri, "Portfolio Choices, Firm Shocks, and Uninsurable Wage Risk," *The Review of Economic Studies*, January 2018, *85* (1), 437–474.

Guerrieri, Veronica, Robert Shimer, and Randall Wright, "Adverse Selection in Competitive Search Equilibrium," *Econometrica*, 2010.

Guiso, Luigi and Luigi Pistaferri, "The Insurance Role of the Firm," *The Geneva Risk and Insurance Review*, 2020, *45* (1), 1–23.

_ , _ , and Fabiano Schivardi, "Insurance within the Firm," *Journal of Political Economy*, 2005, *113* (5), 1054–1087.

_ , _ , and _ , "Credit within the Firm," *The Review of Economic Studies*, January 2013, *80* (1), 211–247.

Halvorsen, Elin, Serdar Ozkan, and Sergio Salgado, "Earnings dynamics and its intergenerational transmission: Evidence from Norway," *Quantitative Economics*, 2022, *13* (4), 1707–1746.

Hopenhayn, Hugo A. and Juan Pablo Nicolini, "Optimal Unemployment Insurance," *Journal of Political Economy*, April 1997, *105* (2), 412–438.

Huggett, Mark, "The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies," *Journal of Economic Dynamics and Control*, September 1993, *17* (5–6), 953–969.

İmrohoroğlu, Ayşe, "Cost of Business Cycles with Indivisibilities and Liquidity Constraints," *Journal of Political Economy*, 1989.

Jolivet, Grégory, Fabien Postel-Vinay, and Jean-Marc Robin, "The empirical content of the job search model: Labor mobility and wage distributions in Europe and the US," *European Economic Review*, 2006, *50* (4), 877–907.

Kaas, Leo, Etienne Lalé, and Nawid Siassi, "Job Ladder and Wealth Dynamics in General Equilibrium," *Econometrica*, 2026.

Kaplan, Greg and Giovanni L. Violante, "A Model of the Consumption Response to Fiscal Stimulus Payments," *Econometrica*, July 2014, *82* (4), 1199–1239.

Kiyotaki, Nobuhiro and Shengxing Zhang, "Intangibles, Inequality and Stagnation," 2024.

Kline, Patrick, Neviana Petkova, Heidi Williams, and Owen Zidar, "Who Profits from Patents? Rent-Sharing at Innovative Firms," *The Quarterly Journal of Economics*, 03 2019, *134* (3), 1343–1404.

Kocherlakota, Narayana R., "Figuring out the impact of hidden savings on optimal unemployment insurance," *Review of Economic Dynamics*, 2004, *7* (3), 541–554.

Krusell, Per, Toshihiko Mukoyama, and Ayşegül Şahin, "Labour-Market Matching with Precautionary Savings and Aggregate Fluctuations," *The Review of Economic Studies*, October 2010, *77* (4), 1477–1507.

Lagakos, David and Guillermo L. Ordoñez, "Which workers get insurance within the firm?," *Journal of Monetary Economics*, 2011, *58* (6), 632–645.

Lentz, Rasmus, "Optimal Employment Contracts with Hidden Search," Working Paper 19988, National Bureau of Economic Research March 2014.

Lise, Jeremy, "On-the-Job Search and Precautionary Savings," *The Review of Economic Studies*, 2013, *80* (3), 1086–1113.

Malgieri, Cedomir and Luca Citino, "Wage Contracts and Financial Frictions," 2025.

Meeuwis, Maarten, Dimitris Papanikolaou, and Lawrence D. W. Schmidt, "Risk Premia, Firm Insurance, and Endogenous Labor Income Risk," Working Paper 35781, National Bureau of Economic Research September 2026.

Menzio, Guido and Shouyong Shi, "Block Recursive Equilibria for Stochastic Models of Search on the Job," *Journal of Economic Theory*, July 2010, *145* (4), 1453–1494.

_ and _, "Efficient Search on the Job and the Business Cycle," *Journal of Political Economy*, June 2011, *119* (3), 468–510.

Moen, Espen R., "Competitive Search Equilibrium," *Journal of Political Economy*, 1997.

Nattinger, Michael B. and Jonathan Rothbaum, "The Effects of Firm Volatility on Workers," 2025.

Oikonomou, Rigas, "Unemployment Insurance with Limited Commitment Wage Contracts and Savings," *BE Journal of Macroeconomics*, 2018.

Petrongolo, Barbara and Christopher A. Pissarides, "Looking into the Black Box: A Survey of the Matching Function," *Journal of Economic Literature*, June 2001, *39* (2), 390–431.

Pissarides, Christopher A., "Consumption and Savings with Unemployment Risk: Implications for Optimal Employment Contracts," *IZA Discussion Paper*, 2004.

Postel-Vinay, Fabien and Jean-Marc Robin, "To match or not to match?: Optimal wage policy with endogenous worker search intensity," *Review of Economic Dynamics*, 2004, *7* (2), 297–330.

Rogerson, William P., "Repeated Moral Hazard," *Econometrica*, January 1985, *53* (1), 69–76.

Shi, Shouyong, "Directed Search for Equilibrium Wage-Tenure Contracts," *Econometrica*, 2009.

Shimer, Robert and Iván Werning, "Liquidity and Insurance for the Unemployed," *American Economic Review*, December 2008, *98* (5), 1922–1942.

Souchier, Martin, "The Pass-through of Productivity Shocks to Wages and the Cyclical Competition for Workers," 2023.

Stevens, Margaret, "Wage-Tenure Contracts in a Frictional Labour Market: Firms' Strategies for Recruitment and Retention," *The Review of Economic Studies*, 04 2004, *71* (2), 535–551.

Talesara, Ishaana and Arthur Wickard, "Monopsony with Dynamic Wage Contracts," 2026.

Thomas, Jonathan and Tim Worrall, "Self-Enforcing Wage Contracts," *Review of Economic Studies*, October 1988, *55* (4), 541–553.

Werning, Iván, "Repeated Moral-Hazard with Unmonitored Wealth: A Recursive First-Order Approach," 2001.

_ , "Optimal Unemployment Insurance with Unobservable Savings," 2002.

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A Theory appendix

A.1 Proof of proposition 1

The optimal contract with transfers when $\delta^{ee} = 0$ solves

$$\begin{aligned}\Pi(s_t) = \max_{w_t,\tau_t^{eu},\tau_t^{ee},c_t,a_{t+1},V(x_{t+1})} & (1-\delta)(1-p(W_t,a_t+\tau_t^{ee}))\left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}[\Pi(s_{t+1})|x_t]}{1+r}\right) \\ & -(1+r)\left[\delta\tau_t^{eu} + (1-\delta)p(W_t,a_t+\tau_t^{ee})\tau_t^{ee}\right]\end{aligned} \tag{A.1}$$

subject to

$$\begin{aligned}&\text{(PK)}: && V_t \leq \delta U(a_t+\tau_t^{eu}) + (1-\delta)\left[W_t + S(W_t, a_t+\tau_t^{ee})\right] \\ &\text{(Budget)}: && c_t + a_{t+1} = (1+r)a_t + w_t \\ &\text{(BC)}: && a_{t+1} \geq 0\end{aligned}$$

Part 1 Consider an allocation that solves the optimal contract (A.1) with a path for assets $\{a_{t+1}(x^t)\}$, where x^t denotes the history of productivity since the beginning of the match. I show that we can construct new paths for wages $\{\tilde{w}_t(x^t)\}$ and transfers $\{\tilde{\tau}_t^{eu}(x^t), \tilde{\tau}_t^{ee}(x^t)\}$ such that the allocations $\{c_t(x^t), p_t(x^t)\}$ remain the same, satisfy all the constraints plus the hand-to-mouth constraint $\tilde{a}_{t+1}(x^t) = 0$ and deliver the same initial value to firms and workers.

Let the new paths of transfers be $\tilde{\tau}_t^i(x^t) = \tau_t^i(x^t) + a_t(x^{t-1})$ for $i \in \text{eu}, \text{ee}$. The new path of wages is $\tilde{w}_t(x^t) = w_t(x^t) + (1+r)a_t(x^{t-1}) - a_{t+1}(x^t)$ in all periods except the first, and $\tilde{w}_t = w_t - a_{t+1}$ in the first period. This wage ensures that the budget constraint is still satisfied when consumption $c_t(x^t)$ remains the same.

First, we show that the worker value remains constant after any history with this new contract. The worker value satisfies

$$\begin{aligned}\tilde{V}_t(x^t) = &\ \delta U(\tilde{\tau}_t^{eu}(x^t)) + (1-\delta)\left[\tilde{W}_t(x^t) + S(\tilde{W}_t(x^t), \tilde{\tau}_t^{ee}(x^t))\right] \\ = &\ \delta U(a_t(x^{t-1}) + \tau_t^{eu}(x^t)) + (1-\delta)\left(\tilde{W}_t(x^t) + S(\tilde{W}_t(x^t), a_t(x^{t-1}) + \tau_t^{ee}(x^t))\right)\end{aligned}$$

with $\tilde{W}_t(x^t) = u(c_t(x^t)) + \beta \mathbb{E}_{x_{t+1}}\left[\tilde{V}_{t+1}(x^{t+1})|x^t\right]$. This gives us a system of two equations for $\tilde{V}_t(x^t), \tilde{W}_t(x^t)$ where the exogenous processes are the same under both contracts so it must be that $\tilde{V}_t(x^t) = V_t(x^t)$ and $\tilde{W}_t(x^t) = W_t(x^t)$ after any x^t. In particular, the initial promised value to the worker V_0 is the same.

It is critical that the worker value is the same after *any* history because the mobility decisions depend on the worker continuation value. In particular, the EE transition rate is the same after any history under both contracts since

$$\tilde{p}_t(x^t) \equiv p(\tilde{W}_t(x^t), \tilde{\tau}_t^{ee}(x^t)) = p(W_t(x^t), a_t(x^{t-1}) + \tau_t^{ee}(x^t)) \equiv p_t(x^t)$$

Then, we show that this new contract achieves the same *initial* value to firms so that their incentives to post vacancies remain unchanged. The firm value after some history will differ but this is irrelevant because the firm has commitment. The firm value satisfies

$$\begin{aligned}\tilde{\Pi}_t(x^t) = &\ (1-\delta)(1-p(\tilde{W}_t(x^t), \tilde{\tau}_t^{ee}(x^t)))\left(x_t - \tilde{w}_t(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x_t\right]}{1+r}\right) \\ &-(1+r)\left[\delta\tilde{\tau}_t^{eu}(x^t) + (1-\delta)p(\tilde{W}_t(x^t), \tilde{\tau}_t^{ee}(x^t))\tilde{\tau}_t^{ee}(x^t)\right] \\ = &\ (1-\delta)(1-p(W_t(x^t), a_t + \tau_t^{ee}(x^t)))\left(x_t - w_t(x^t) + a_{t+1}(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x_t\right]}{1+r}\right) \\ &-(1+r)\left[\delta\tau_t^{eu}(x^t) + (1-\delta)p(W_t(x^t), a_t + \tau_t^{ee}(x^t))\tau_t^{ee}(x^t)\right] \\ &-(1+r)a_t(x^{t-1})\end{aligned}$$

where we used that $\tilde{W}_t(x^t) = W_t(x^t)$. Finally, we can guess and verify that $\tilde{\Pi}_t(x^t) = \Pi_t(x^t) - (1+r)a_t(x^{t-1})$ for all x^t. Now plug this in the problem of new entrants to verify that the initial firm value is the same

$$\begin{aligned}\tilde{\Pi}_0(v, a_t) = &\ 1 - \tilde{w}_t + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})\right]}{1+r} \\ = &\ 1 - w_t + a_{t+1} - a_{t+1} + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})\right]}{1+r} \\ = &\ 1 - w_t + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})\right]}{1+r} \\ = &\ \Pi_0(v, a_t)\end{aligned}$$

Part 2 I prove part 2 under the assumption that productivity is constant within matches so I will denote the wage as w_t, where t denotes time. I also assume that the borrowing constraint is finite to ensure transversality but sufficiently relaxed so that assets can be chosen freely. Denote the allocation achieved with transfers as $\{c_t, p_t\}$ and the corresponding transfers as $\{\tau_t^{eu}, \tau_t^{ee}\}$. We assume that this contract satisfies $a_{t+1} = 0$ for all t, which is without loss of generality from part 1.

We now construct a new contract consisting of paths for the wages, the assets and the transfers $\{\tilde{w}_t, \tilde{a}_{t+1}, \tilde{\tau}_t^{eu}, \tilde{\tau}_t^{ee}\}$ such that

a) firms and workers can commit to all transfers *but one*, denoted $\tilde{\tau}_t^i$ where $i \in \{\text{eu}, \text{ee}\}$, so $\tilde{\tau}_t^i = 0$ for all t,

b) the promise keeping constraint and the budget constraint are satisfied for all t,

c) the worker consumption $\tilde{c}_t$, EE transition rate $\tilde{p}_t$, and values $\tilde{V}_t, \tilde{W}_t$ are the same for all t,

d) the initial firm value $\tilde{\Pi}_0(v, a)$ is the same.

This contract achieves the same firm value as the contract in the relaxed problem where τ_t^i is unrestricted so it must be optimal when $\tau_t^i = 0$. If this new contract exists, it shows that we can implement the same allocation with a contract that satisfies $\tau_t^i = 0$ provided that workers can trade assets a_{t+1}.

Construct the new contract $\{\tilde{w}_t, \tilde{a}_{t+1}, \tilde{\tau}_t^{eu}, \tilde{\tau}_t^{ee}\}$ using

$$\tilde{w}_t = w_t - (1+r)\tilde{a}_t + \tilde{a}_{t+1}$$

$$\tilde{a}_{t+1} = \tau_{t+1}^i$$

$$\tilde{\tau}_t^i = 0$$

$$\tilde{\tau}_t^j = \tau_t^j - \tilde{a}_t \qquad \text{for all } j \neq i$$

except for the first period where $\tilde{w}_t = w_t + \tilde{a}_{t+1}$. At $t = 0$, there are no EE or EU transitions so the transfers τ_0^{eu}, τ_0^{ee} are irrelevant and $\tilde{a}_0 = a_0$. The consumption of the worker implied by this path is unchanged since

$$\tilde{c}_t = (1+r)\tilde{a}_t + \tilde{w}_t - \tilde{a}_{t+1} = w_t = c_t$$

for all periods except the first, and

$$\tilde{c}_t = (1+r)\tilde{a}_t + \tilde{w}_t - \tilde{a}_{t+1} = (1+r)a_t + w_t = c_t$$

for the first period.

We now show that the worker value remains constant after any history with this new contract. The worker value satisfies

$$\begin{aligned} \tilde{V}_t = &\quad \delta U(\tilde{a}_t + \tilde{\tau}_t^{eu}) + (1-\delta)\left[\tilde{W}_t + S(\tilde{W}_t, \tilde{a}_t + \tilde{\tau}_t^{ee})\right] \\ = &\quad \delta U(\tau_t^{eu}) + (1-\delta)\left[\tilde{W}_t + S(\tilde{W}_t, \tau_t^{ee})\right] \end{aligned}$$

with $\tilde{W}_t = u(c_t) + \beta \tilde{V}_{t+1}$. This gives us a system of two equations for $\tilde{V}_t, \tilde{W}_t$ where the exogenous processes are the same under both contracts so it must be that $\tilde{V}_t = V_t$ and $\tilde{W}_t = W_t$ for all t. In particular, the initial promised value to the worker V_0 is the same. It also follows that the EE transition rate is the same after any history under both contracts since

$$\tilde{p}_t \equiv p(\tilde{W}_t, \tilde{a}_t + \tilde{\tau}_t^{ee}) = p(W_t, \tau_t^{ee}) \equiv p_t$$

Finally, we show that the firm achieves the same initial value with this new contract. The firm value at time t satisfies

$$\begin{aligned} \tilde{\Pi}_t = &\quad (1-\delta)(1 - p(\tilde{W}_t, \tilde{a}_t + \tilde{\tau}_t^{ee}))\left(1 - \tilde{w}_t + \tfrac{\tilde{\Pi}_{t+1}}{1+r}\right) \\ &\qquad -(1+r)\left[\delta \tilde{\tau}_t^{eu} + (1-\delta)p(\tilde{W}_t, \tilde{a}_t + \tilde{\tau}_t^{ee})\tilde{\tau}_t^{ee}\right] \\ = &\quad (1-\delta)(1 - p(W_t, \tau_t^{ee}))\left(1 - w_t - \tilde{a}_{t+1} + \tfrac{\tilde{\Pi}_{t+1}}{1+r}\right) \\ &\qquad -(1+r)\left[\delta \tau_t^{eu} + (1-\delta)p(W_t, \tau_t^{ee})\tau_t^{ee}\right] \\ &\qquad +(1+r)\tilde{a}_t \end{aligned}$$

where we used that $\tilde{W}_t = W_t$. Finally, we can guess and verify that $\tilde{\Pi}_t = \Pi_t + (1+r)\tilde{a}_t$ for all t. Now plug this in the problem of new entrants to verify that the initial firm value is the same

$$\begin{aligned}\tilde{\Pi}_0(v, a_t) &= 1 - \tilde{w}_t + \tfrac{\tilde{\Pi}_{t+1}}{1+r} \\ &= 1 - w_t - \tilde{a}_{t+1} + \tilde{a}_{t+1} + \tfrac{\Pi_{t+1}}{1+r} \\ &= 1 - w_t + \tfrac{\Pi_{t+1}}{1+r} \\ &= \Pi_0(v, a_t)\end{aligned}$$

A.2 Consumption growth condition (5)

Consider the optimal contract (3) and denote the Lagrange multipliers on the constraints as $\eta_t, \mu_t, \zeta_t, \lambda_t$. The optimality conditions are

$$\begin{aligned}w_t: &\quad (1-\delta)(1-\delta^{ee})(1-p(W_t,a_t)) = \mu_t \\ c_t: &\quad \lambda_t u'(c_t) = \mu_t \\ V(x_{t+1}): &\quad (1-\delta)(1-\delta^{ee})(1-p(W_t,a_t))\tfrac{\Pi_V(s_{t+1})}{1+r} + \beta\lambda_t = 0 \\ W_t: &\quad -(1-\delta^{ee})p_W(W_t,a_t)\left(x_t - w_t + \tfrac{\mathbb{E}_{x_{t+1}}[\Pi(s_{t+1})|x_t]}{1+r}\right) \\ &\qquad +\eta_t\left[\delta^{ee} + (1-\delta^{ee})(1-p(W_t,a_t))\right] - \tfrac{\lambda_t}{(1-\delta)} = 0 \\ a_{t+1}: &\quad (1-\delta)(1-\delta^{ee})(1-p(W_t,a_t))\tfrac{\mathbb{E}_{x_{t+1}}[\Pi_a(s_{t+1})|x_t]}{1+r} = \mu_t - \zeta_t\end{aligned}$$

and the envelope conditions are

$$\begin{aligned}V_t: &\quad \Pi_V(s_t) = -\eta_t \\ a_t: &\quad \Pi_a(s_t) = -(1-\delta)(1-\delta^{ee})p_a(W_t,a_t)\left(x_t - w_t + \tfrac{\mathbb{E}_{x_{t+1}}[\Pi(s_{t+1})|x_t]}{1+r}\right) \\ &\qquad +\eta_t\left(\delta U'(a_t) + (1-\delta)(1-\delta^{ee})S_a(W_t,a_t)\right) + \mu_t(1+r)\end{aligned}$$

Combine the optimality conditions with respect to $V(x_{t+1})$ and W_t in the optimal contract, and the envelope condition with respect to V_t to get

$$\frac{\eta_{t+1} - \eta_t}{\beta(1+r)} = -\frac{p_W(W_t,a_t)}{1-p(W_t,a_t)}\left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}[\Pi(s_{t+1})|x_t]}{1+r}\right) + \frac{\eta_t\Phi(W_t,a_t)}{\beta(1+r)}$$

with $\Phi(W_t,a_t) \equiv \beta(1+r)\left[1 + \frac{\delta^{ee}}{1-\delta^{ee}}\frac{1}{1-p(W_t,a_t)}\right] - 1$. Now combine the first-order conditions for c_t, w_t and $V(x_{t+1})$ and the envelope condition for V_t to get

$$\eta_{t+1} = \frac{\beta(1+r)}{u'(c_t)} \tag{A.2}$$

and therefore

$$\frac{1}{u'(c_t)} - \frac{1}{u'(c_{t-1})} = -\frac{p_W(W_t,a_t)}{1-p(W_t,a_t)}\left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}[\Pi(s_{t+1})|x_t]}{1+r}\right) + \frac{\Phi(W_t,a_t)}{u'(c_{t-1})}$$

which is equation (5).

A.3 Proof of proposition 2

First, combine the first-order condition for a_{t+1} with the envelope condition for a_t to get

$$\begin{aligned} 1+r \geq\ & \mathbb{E}_{x_{t+1}}\left[\eta_{t+1}\left(\delta U'(a_{t+1}) + (1-\delta)(1-\delta^{ee})S_a(W_{t+1},a_{t+1})\right)|x_t\right] \\ & + \mathbb{E}_{x_{t+1}}\left[(1-\delta)(1-\delta^{ee})(1-p(W_{t+1},a_{t+1}))(1+r)|x_t\right] \\ & - \mathbb{E}_{x_{t+1}}\left[(1-\delta)(1-\delta^{ee})p_a(W_{t+1},a_{t+1})\left(x_{t+1}-w_{t+1}+\frac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}),a_{t+2},x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\right] \end{aligned} \tag{A.3}$$

with equality if the borrowing constraint does not bind.

Next, combine the first-order condition for c_t and the envelope condition of the unemployed workers to get

$$U'(a_t) = (1+r)u'(c_t^u) \tag{A.4}$$

where c_t^u denotes the consumption of the unemployed worker.

We need to derive an expression for $S_a(W_{t+1},a_{t+1})$. From the envelope condition on the search problem of workers (2), we get

$$S_a(W_t,a_t) \equiv \kappa\left(v(W_t,a_t)-W_t\right)\partial_a\lambda_w(v(W_t,a_t),a_t)$$

We now derive an expression for $\partial_a\lambda_w(v(W_t,a_t),a_t)$ from the free entry condition of firms. Combining the first-order conditions and envelope conditions in the problem of new entrants gives

$$\partial_v\Pi_0(v,a_t) = -\frac{1}{u'(c_t)}$$

$$\partial_a\Pi_0(v,a_t) = 1+r$$

where c_t represents the consumption of the worker at the new job. Now consider the free entry condition

$$\lambda_f(v,a) = \frac{k}{\Pi_0(v,a)}$$

Differentiating this expression with respect to v and a gives

$$\partial_a\lambda_f(v,a) = -\lambda_f(v,a)\frac{1+r}{\Pi_0(v,a)}$$

$$\partial_v\lambda_f(v,a) = \lambda_f(v,a)\frac{1}{\Pi_0(v,a)u'(c_t)}$$

Taking the ratio gives

$$\partial_a\lambda_f(v,a) = -\partial_v\lambda_f(v,a)(1+r)u'(c_t)$$

With a constant returns to scale matching function, we can express the job finding rate as $\lambda_w(v,a) = f(\lambda_f(v,a))$. Therefore,

$$\partial_a\lambda_w(v,a) = -\partial_v\lambda_w(v,a)(1+r)u'(c_t) \tag{A.5}$$

We can use this expression to rewrite $S_a(W_t,a_t)$ as

$$S_a(W_t,a_t) = -(1+r)u'(c_t^{ee})\kappa\left(v(W_t,a_t)-W_t\right)\partial_v\lambda_w(v(W_t,a_t),a_t)$$

where c_t^{ee} is the consumption of the worker during the first period after an endogenous EE sepa-

ration. We can further simplify this term using the first-order condition of the search problem

$$\lambda_w(v_t, a_t) + \partial_v \lambda_w(v_t, a_t)\,(v_t - W_t) = 0$$

and get

$$S_a(W_t, a_t) = (1+r)u'(c_t^{ee})\kappa\lambda_w(v(W_t, a_t), a_t) = (1+r)u'(c_t^{ee})p(W_t, a_t) \tag{A.6}$$

Combine equations (A.2), (A.3), (A.4) and (A.6) to get

$$\begin{aligned} u'(c_t) \geq\ & \mathbb{E}_{x_{t+1}}\left[\beta(1+r)\left(\delta u'(c_{t+1}^u) + (1-\delta)(1-\delta^{ee})p(W_{t+1}, a_{t+1})u'(c_{t+1}^{ee})\right)|x_t\right] \\ & +(1-\delta)(1-\delta^{ee})\mathbb{E}_{x_{t+1}}\left[(1-p(W_{t+1}, a_{t+1}))u'(c_t)|x_t\right] \\ & -(1-\delta)(1-\delta^{ee})u'(c_t)\mathbb{E}_{x_{t+1}}\left[\tfrac{p_a(W_{t+1}, a_{t+1})}{1+r}\left(x_{t+1} - w_{t+1} + \tfrac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\right] \end{aligned}$$

For the final step, rewrite the consumption growth condition (5) evaluated at $t+1$ as

$$\begin{aligned} u'(c_t) = & u'(c_{t+1})\beta(1+r)\left[1 + \tfrac{\delta^{ee}}{1-\delta^{ee}}\tfrac{1}{1-p(W_{t+1}, a_{t+1})}\right] \\ & -u'(c_t)u'(c_{t+1})\tfrac{p_W(W_{t+1}, a_{t+1})}{1-p(W_{t+1}, a_{t+1})}\left(x_{t+1} - w_{t+1} + \tfrac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right) \end{aligned}$$

We can replace $u'(c_t)$ in the second line by this expression and use the definition of $\mathcal{W}_t$ to get

$$\begin{aligned} \tfrac{u'(c_t)}{\beta(1+r)} \geq\ & \delta u'(c_{t+1}^u) + (1-\delta)\delta^{ee}\mathbb{E}_{x_{t+1}}\left[u'(c_{t+1})|x_t\right] \\ & +(1-\delta)(1-\delta^{ee})\mathbb{E}_{x_{t+1}}\left[p(W_{t+1}, a_{t+1})u'(c_{t+1}^{ee})|x_t\right] \\ & +(1-\delta)(1-\delta^{ee})\mathbb{E}_{x_{t+1}}\left[(1-p(W_{t+1}, a_{t+1}))u'(c_{t+1})|x_t\right] + \mathcal{W}_t \end{aligned}$$

which is equation (7).

A.4 Optimality conditions for unemployed workers

The optimality condition with respect to v_{t+1} leads to the standard optimality condition in directed search models

$$\lambda_w(v_{t+1}, a_{t+1}) = -\partial_v \lambda_w(v_{t+1}, a_{t+1})\left[v_{t+1} - U(a_{t+1})\right]$$

It states that workers equate the benefits of search in markets with higher values v_{t+1} to the cost in terms of decreased match probability.

Combining the optimality conditions with respect to c_t and a_{t+1} and the envelope condition gives

$$u'(c_t) \geq \beta\partial_a \lambda_w(v_{t+1}, a_{t+1})\left[v_{t+1} - U(a_{t+1})\right] + \beta(1-\lambda_w(v_{t+1}, a_{t+1}))(1+r)u'(c_{t+1})$$

Finally, we can use equation (A.5) and the optimality condition for v_{t+1} to get

$$u'(c_t) \geq \beta(1+r)\left[\lambda_w(v_{t+1}, a_{t+1})u'(c_{t+1}^{ue}) + (1-\lambda_w(v_{t+1}, a_{t+1}))u'(c_{t+1})\right]$$

which is a standard Euler equation.

B Quantitative results appendix

B.1 Matched employer-employee data

I use administrative data provided by CASD in France between 2005 and 2019. My analysis relies on two main files:

a) the panel version of the "DADS tous salariés" database, containing detailed information about employment history for 1/12th of the French population;

b) "FARE" database, with annual information about firm balance sheets and income statements for the entire private sector, except agricultural firms

Sample selection From the FARE file on firms, I exclude firms belonging to the public administration and non-profit organizations. From the DADS file, I keep prime-age workers (25 to 55 years old) with full-time jobs and permanent contracts (CDI). I focus on relatively stable jobs because I study the problem of worker retention, and it would not fit very well the case of temporary contracts (CDD) since they usually end after a short period of time. The variable on permanent contracts only starts in 2005 so this determines the start of my sample period. I restrict my sample to workers and firms who are in the panel for at least 3 years and to firms with at least 3 employees. I drop firms with negative or missing labor productivity. I remove outliers below and above the 1st and 99th percentiles for productivity growth, wage and wage growth.

Definition of wages and labor productivity I measure labor productivity annually as value added per worker. I define wages as total labor earnings in a year divided by the number of days worked at that firm. Labor earnings are net of payroll taxes but gross of income taxes. They include regular wages, overtime pay, bonuses and even payment in kind but exclude stock options, which are not measured.

Definition of labor market flows Identifying EE transitions is challenging because workers sometimes hold multiple jobs at the same time. For this reason, I first identify the main job of a worker defined as the job with the earliest start date. I drop jobs that lasted less than 35 hours in a year and all main jobs if they end up accounting for less than 50% of total earnings from simultaneous jobs. I also drop individuals with more than 5 jobs in a given year.

I use the exact start and end dates of jobs to identify a job separation. An EE transition occurs if the new job starts 18 days or less after the previous job ends. This leaves a little bit of room for workers who take 2 weeks of holidays in between jobs. The risk is that it might also include workers who transit through unemployment for just 2 weeks and find a new job quickly. Note however that France is a country in which the job finding rate is fairly low (I estimate 20% per quarter) so most likely this risk is minimal. I also count cases where the new and old jobs overlap for some time as EE transitions.

When a worker separates from their previous job and does not make an EE separation, I define it as a separation into non-employment.

I compute the duration of non-employment as the number of months until a worker reappears in my sample, conditional on the worker reappearing. By conditioning on whether a worker ever comes back in my sample I sort out workers who leave the labor force permanently (e.g. retirement, death). I only estimate this moment for the period 2005-2015 so that workers have plenty of time to come back before my sample ends in 2019.

B.2 Estimation details

I now provide details on the model estimation.

Estimation in two steps The estimation has two steps. First, I estimate the process for productivity. Second, I estimate the remaining model parameters.

To estimate the process for productivity, I simulate productivity at the quarterly frequency, aggregate the data at the annual level and then add measurement errors. I then estimate the volatility and persistence of productivity and the volatility of measurement errors by comparing the variance of productivity growth and autocorrelation of orders 1 and 2 in the model-generated data and the administrative data.

To estimate the remaining parameters, I simulate a panel of workers at the quarterly frequency using the policy functions of the model and the estimated parameters for the productivity process. I then aggregate at the annual level to create an annual panel of workers, similar to the data. I use this panel to compute the moments used in the estimation of the remaining parameters.

Pass-through and MPC Denote the growth rates for productivity, wages and consumption by $\Delta \log x_t$, $\Delta \log w_t$ and $\Delta \log c_t$ respectively. Following Guiso et al. (2005), the pass-through is measured in the data and the model using the following estimator

$$\frac{\text{Cov}(\Delta \log w_t, \Delta \log x_{t-1} + \Delta \log x_t + \Delta \log x_{t+1})}{\text{Cov}(\Delta \log x_t, \Delta \log x_{t-1} + \Delta \log x_t + \Delta \log x_{t+1})} \tag{A.7}$$

which helps account for transitory measurement errors.

The MPC out of earnings shocks is measured in a similar way using

$$\frac{\text{Cov}(\Delta \log c_t, \Delta \log x_{t-1} + \Delta \log x_t + \Delta \log x_{t+1})}{\text{Cov}(\Delta \log w_t, \Delta \log x_{t-1} + \Delta \log x_t + \Delta \log x_{t+1})}$$

where the moving average of productivity growth is effectively used as an instrument for earnings growth. There are two reasons to use this estimator as opposed to a simpler OLS estimator of consumption growth on wage growth. The first is that wages change over time even in the absence of shocks because of tenure profiles, and the consumption response to changes in earnings over time might be very different than the MPC out of earnings shocks. Second, using the moving average of productivity growth helps filter out transitory measurement errors on productivity.

The model-implied MPC is then compared to the estimates of Blundell et al. (2008). They report estimates of MPC of 64% annually for permanent shocks and close to 0% for transitory shocks. For college workers, they estimate an MPC of 42%. In my sample, shocks are quite persistent ($\rho = 0.926$ quarterly), though not entirely permanent, and workers have stronger attachment to labor markets because they hold permanent contracts with full-time jobs. For this reason, I use an estimate of 40% for my baseline estimate of MPC but also consider alternative specifications targeting an MPC of 50% and 30% in section B.3. The quantitative results from section 4 are similar across these specifications, even though workers receive different amounts of insurance inside and outside the firm.

Wage tenure profile The wage tenure profile is estimated in the data using dummies for tenure (less than 1 year, 1 to 3 years, 3 to 5, 5 to 10, 10 to 15, 15 to 20 and equal or higher to 20 years) together with the same dummies for worker experience (age minus 20). In the model-generated

	Baseline	Persistent meas. errors	Low MPC	High MPC	Low tenure profile
Cost of posting vacancy k	3.33	2.3	4.24	2.74	3.51
Exogenous EE transition rate δ^{ee}	0.017	0	0.016	0.017	0.016
Separation rate δ	0.018	0.018	0.018	0.018	0.018
Flow unemployment value b	0.51	0.6	0.42	0.57	0.5
On-the-job search efficiency κ	0.07	0.376	0.068	0.077	0.071
Relative discount factor $\beta(1+r)$	0.983	0.98	0.983	0.983	0.981
Volatility of productivity σ	0.0864	0.0871	0.0864	0.0864	0.0864
Persistence of productivity ρ	0.926	0.734	0.926	0.926	0.926
Volatility of meas. error σ^{meas}	0.107	0.087	0.107	0.107	0.107
Volatility of persistent meas. error	0	0.114	0	0	0

Note: estimated parameter values in alternative specifications of the model.

Table A.1: Estimated parameters in alternative models

data, the tenure profile is estimated using the same dummies for tenure, but without the experience dummies because I consider a stationary equilibrium. The coefficient reported in table 1 is the coefficient on tenure equal to or higher than 20 years.

One concern with this approach is that wages could grow with tenure because of human capital accumulation that is not in the model. If this human capital is not transferable across jobs, then it would not be captured in my dummies for worker experience. To account for this possibility, I estimate in section B.3 a version of the model that targets a tenure profile that is half as large as the one I measure in the data. I find that my results are robust to this target.

Labor market transitions Unemployment duration and transitions out of employment (EE and EU) are computed similarly in the model and the data.

B.3 Alternative model specifications

In the data, the pass-through of productivity shocks to wages is small. This creates a tension with the model with assets where the pass-through tends to be larger. In the model of section 2, introducing exogenous shocks to EE mobility resolved this tension. In this section, I provide an alternative solution based on persistent measurement errors and show that the main quantitative results from section 4 about crowding out are robust to this alternative specification. I also consider alternative targets for the marginal propensity to consume and the wage tenure profile.

Tension between model and data To illustrate the tension between model and data, I first estimate a model without exogenous EE mobility and without persistent measurement error to show that the pass-through tends to be larger in the model than in the data. To estimate this model, I do not target the pass-through. The estimated search efficiency of employed workers κ is now much larger relative to section 4.1 (0.39 vs 0.07) because all EE transitions are endogenous. The wage pass-through implied by the model is 27.9%, which is much higher than the one estimated in the data (5.8%). The reason why the pass-through is so large in the model is that firms seek to influence worker mobility. To see this, consider the response of EE transitions to a productivity

	Baseline	Persistent meas. errors	Low MPC	High MPC	Low tenure profile
Wage growth p90 vs p10	0.68	0.13	1.29	0.41	0.42
Cons. growth p90 vs p10	-0.1	-0.08	-0.16	-0.07	-0.07
Wage pass-through p80 vs p10	5.31	6.09	3.95	4.84	4.48
Cons. pass-through p80 vs p10	-2.43	-0.23	-1.73	-3.24	-1.6
Average wage p90 vs p10	0.85	0.16	1.46	0.52	0.66
Wage growth $\underline{a} = -1$ vs $\underline{a} = 0$	1.56	1.35	1.5	1.64	1.31
Cons. growth $\underline{a} = -1$ vs $\underline{a} = 0$	-0.13	-0.3	-0.11	-0.15	-0.12
Wage pass-through $\underline{a} = -1$ vs $\underline{a} = 0$	0.63	0.53	0.46	0.73	0.54
Cons. pass-through $\underline{a} = -1$ vs $\underline{a} = 0$	-0.08	-0.14	-0.04	-0.15	-0.04
Average wage $\underline{a} = -1$ vs $\underline{a} = 0$	1.08	0.46	1.19	0.94	1.15
Wage growth $h = 0.5$ vs $h = 0$	0.21	0.51	0.97	-0.06	0.41
Cons. growth $h = 0.5$ vs $h = 0$	-0.02	-0.06	-0.05	0.01	-0.02
Wage pass-through $h = 0.5$ vs $h = 0$	-2.36	-2.78	-1.79	-2.16	-2.58
Cons. pass-through $h = 0.5$ vs $h = 0$	0.2	0.43	0.03	0.36	0.17
Average wage $h = 0.5$ vs $h = 0$	0.19	0.06	0.58	0.06	0.27

Note: this table summarizes the main quantitative results from section 4 in the baseline model of section 2 and in alternative models with persistent measurement errors, and targeting different estimates of MPC or tenure profile for wages. All numbers are in percentage point differences, except for average wage that is in percent. h denotes the proportional increase in unemployment benefits.

Table A.2: Quantitative results across alternative models

shock, measured by replacing wages with an indicator for EE mobility in equation (A.7). This moment is -13.0% in the model without exogenous EE mobility, compared to -1.1% in the data. These results show that, in a model without exogenous EE transitions, wage contracts have a great influence on worker mobility, which induces firms to set a high degree of pass-through.

The model of section 2 resolves this tension with the introduction of exogenous EE transitions. These transitions, sometimes called godfather shocks, are common features of search models (e.g. Jolivet et al. 2006). In my model, exogenous transitions reduce the influence of wage contracts on EE mobility. As a result, firms seek to provide more insurance to workers when they design wage contracts, and to optimize worker retention less. This is why the optimal pass-through in section 2 is in line with the data. The pass-through of productivity shocks to the EE transition probability of workers is -0.7% in this model, thus broadly consistent with the data. This suggests that the model with exogenous EE transitions captures well how wage contracts influence worker mobility.

Alternative model based on measurement errors An alternative solution for the tension between model and data is that the pass-through estimated in the data does not reflect the true pass-through of shocks to wages because of persistent measurement errors, which leads to attenuation bias. The pass-through I estimate is consistent with the existing literature (see Guiso and Pistaferri, 2020 for a review) but this literature assumes that measurement errors are transitory and not persistent. In fact, recent estimates of pass-through using exogenous shocks to firm profitability tend to find pass-through estimates that are larger in magnitude (e.g. Kline, Petkova,

Williams and Zidar, 2019)[17]. To account for this possibility, I estimate the model without exogenous EE transitions ($\delta^{ee} = 0$) but with permanent measurement errors to productivity in addition to transitory measurement errors. Specifically, observed annual log productivity is now the sum of true log productivity, a transitory measurement error and a random walk component. I estimate the size of measurement errors by targeting the pass-through observed in administrative data. In this exercise, I leave the tenure profile untargeted and set $b = 0.6$. Column 2 of table A.1 reports the estimated parameter values from this exercise. The pass-through of productivity shocks to wages, measured *without measurement errors*, is 16.2%, which is consistent with existing estimates based on identified productivity shocks and about 3 times larger than my pass-through estimates from administrative data. The response of EE transitions to productivity shocks, measured with measurement errors, is -0.6%, thus broadly consistent with the data.

In this model, workers receive less insurance inside firms because the pass-through to wages is larger. However, table A.2 shows that the main quantitative predictions from section 4 go through. For the heterogeneity in pass-through across the wealth distribution, the table focuses on the p80-p10 difference because the pass-through falls with assets at the very top of the distribution. The reason is that wealthy workers also happen to be at the top of the job ladder where firm profits are small, which attenuates the desire of firms to retain these workers.

To summarize, the main crowding-out results hold under both explanations for the low measured pass-through.

Targeting different estimates of MPC The baseline model targets an annual MPC of 40%. I now consider two alternative specifications targeting an MPC of 30% and 50%. The estimated parameters are shown in columns 3 and 4 of table A.1 and the implied quantitative results are shown in table A.2. Across these alternative specifications, the results from section 4 are broadly consistent.

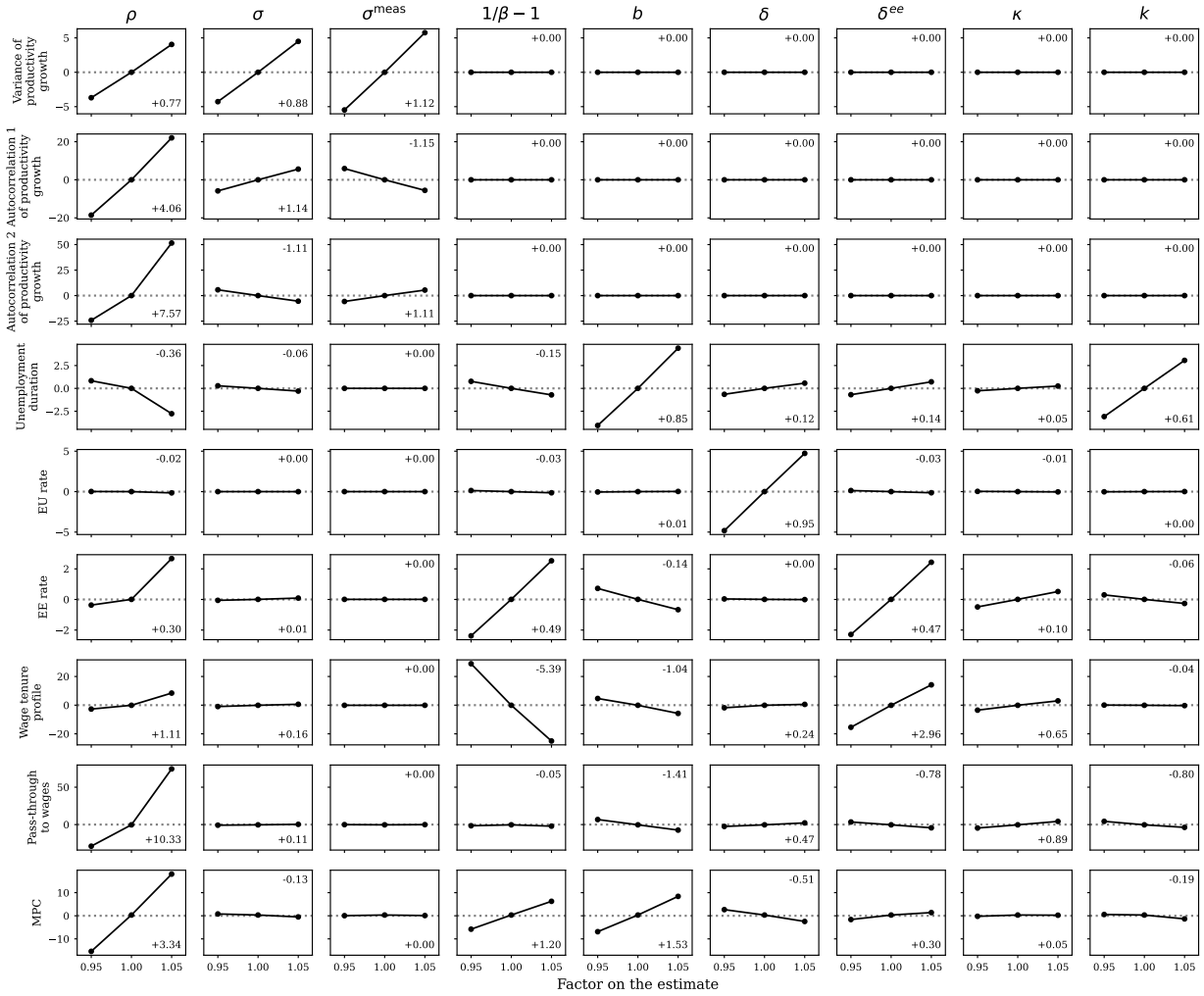
Targeting a smaller tenure profile for wages The baseline model targets a tenure profile for wages of 10.3%, as estimated in the data. I consider an alternative specification targeting a tenure profile of 5.15% to account for the possibility that some of the observed growth in wages can be due to human capital accumulation that is not transferable across jobs. The estimated parameters are shown in column 5 of table A.1 and the implied quantitative results are shown in table A.2. The results are robust to this modification.

B.4 Additional estimation results

Figure A.1 shows how different parameters influence each moment at the estimated parameter values whereas figure A.2 shows the sensitivity of estimated parameters to the value of moments following Andrews, Gentzkow and Shapiro (2017). The Jacobian of the moments with respect to the parameters has full column rank at the estimated parameters (condition number 36.5), so the model is locally identified.

[17] There is a trade-off between internal validity and external validity. The statistical approach of Guiso et al. (2005) uses the universe of firms and workers so the estimates are representative of labor markets in general but they assume that changes in productivity are correctly measured and exogenous. By contrast, existing estimates based on identified shocks to firm productivity often focus on specific samples that might not be representative of labor markets at large.

Figure A.1: Sensitivity of moments to estimated parameters



Note: the figure shows the Jacobian of moments to parameters evaluated at the estimated parameter values.

Figure A.2: Sensitivity of estimated parameters to moments

	Variance of productivity growth	Autocorrelation 1 of productivity growth	Autocorrelation 2 of productivity growth	Unemployment duration	EU rate	EE rate	Wage tenure profile	Pass-through to wages	MPC
ρ	0.00	0.08	0.09	0.00	0.00	0.00	0.00	0.00	0.00
σ	0.50	0.29	-0.21	0.00	0.00	0.00	0.00	0.00	0.00
σ^{meas}	0.50	-0.28	0.10	0.00	0.00	0.00	0.00	0.00	0.00
$1/\beta - 1$	0.00	0.01	0.00	0.05	0.01	0.79	-0.12	0.00	-0.04
b	0.04	-0.12	-0.17	0.12	0.27	-0.72	0.06	0.01	0.57
δ	0.00	0.00	0.00	0.00	1.05	0.06	0.00	0.00	0.00
δ^{ee}	-0.01	0.11	0.12	-0.14	0.13	1.06	0.12	-0.20	0.07
κ	-0.01	-0.86	-0.87	1.34	-0.53	0.59	0.06	0.93	0.21
k	0.00	0.29	0.31	1.39	-0.56	0.88	-0.14	-0.04	-0.82

Note: the figure shows the inverse of the Jacobian of moments to parameters evaluated at the estimated parameter values.

C Hidden assets appendix

C.1 Hidden savings

I solve the optimal contract under the assumption that the savings decision of workers is private information using the first-order approach (Werning, 2001, Ábrahám and Pavoni, 2008), which means that the Euler equation is now an additional constraint in problem (3). The challenge is that the problem must be solved with high numerical accuracy because the solutions with observable savings and hidden savings are so close. For this reason, I focus on a simplified model with constant productivity to reduce the state space. I also keep the equilibrium EE mobility rate and value $p(W,a), S(W,a)$, and the value from unemployment $U(a)$ from the solution of the problem with observable assets.

The optimal contract with hidden savings solves

$$\Pi(V_t, a_t, u'(c_{t-1})) = \max_{w_t, c_t, a_{t+1}, V_{t+1}} (1-\delta)(1-\delta^{ee})(1-p(W_t,a_t)) \left(1 - w_t + \frac{\Pi(V_{t+1}, a_{t+1}, u'(c_t))}{1+r}\right)$$

subject to

$$\begin{aligned}
&\text{(PK)}: && V_t \leq \delta U(a_t) + (1-\delta)\left[\delta^{ee} W_t + (1-\delta^{ee})\left(W_t + S(W_t,a_t)\right)\right] \\
&\text{(Budget)}: && c_t + a_{t+1} = (1+r)a_t + w_t \\
&\text{(BC)}: && a_{t+1} \geq 0 \\
&\text{(Euler)}: && u'(c_{t-1}) \geq \beta\delta U'(a_t) + \beta(1+r)(1-\delta)\delta^{ee} u'(c_t) \\
& && \quad + \beta(1-\delta)(1-\delta^{ee})\left[(1-p(W_t,a_t))(1+r)u'(c_t) + S_a(W_t,a_t)\right]
\end{aligned}$$

where we recall that $W_t = u(c_t) + \beta V_{t+1}$. The last constraint holds with equality when $a_t > 0$. In the initial period, the optimal contract solves the same problem but without $u'(c_{t-1})$ as a state variable, with $\delta = \delta^{ee} = p_t = S_t = 0$ and without the Euler equation as a constraint.

I compute the difference between the paths of wages, consumption, assets and EE transition rate in the model with hidden savings and the model with observable savings for a worker hired from unemployment. I find that the differences are respectively of the order of 10^{-5}, 10^{-5}, 10^{-3} and 10^{-6}. I also check ex-post that the job finding rate implied by the optimal contract computed under the assumption of hidden savings is similar to the one computed with observable savings (difference of 10^{-6}). Therefore, these results show that the solutions to the model with hidden savings or observable savings are virtually identical.

C.2 Hidden initial assets

I consider a 2-period version of my model ($t = 0, 1$) where, for simplicity, I assume constant productivity, no exogenous EE transitions and $\beta = 1 + r = 1$.

Labor markets at $t = 1$ The free entry condition at $t = 1$ is

$$0 \geq -k + \lambda_{1f}(v,a)\left[1 - w(v,a)\right]$$

with equality in active markets. The wage satisfies the promise keeping constraint $v = u(w(v,a) + a)$. $\lambda_{1f}(v,a_1)$ denotes the vacancy filling rate at $t = 1$. Workers' assets do not influence the firms'

profits, except through wages, so workers have no incentives to misreport assets to their new employers.

Worker values at $t = 0$ Consider a worker who did not find a job at $t = 0$. This worker has initial asset a_0, receives b and can search for a job at $t = 1$. The value of this worker at the end of $t = 0$ satisfies

$$U_1(a_0) = \max_{a_1 \geq 0, v} u(b + a_0 - a_1) + u(b + a_1) + \lambda_{1w}(v, a_1)(v - u(b + a_1))$$

where $\lambda_{1w}(v, a_1)$ denotes the job finding rate at $t = 1$.

Consider a worker who found a job at $t = 0$. This worker has initial asset a_0, receives wages $w_0, w_1$and can search for a job at $t = 1$. The value of this worker at the end of $t = 0$ satisfies

$$\max_{a_1 \geq 0, v_1} V_1(a_0, a_1, w_0, w_1, v_1)$$

where

$$V_1(a_0, a_1, w_0, w_1, v_1) = u(w_0 + a_0 - a_1) + \delta u(b + a_1) + (1 - \delta)\left[u(w_1 + a_1) + \kappa\lambda_{1w}(v_1, a_1)(v_1 - u(w_1 + a_1))\right]$$

Optimal contract at $t = 0$ In section 2, we expressed the optimal contract as maximizing profits subject to a promise keeping constraint for the worker. With hidden asset a_0, it is convenient to write it instead as maximizing the worker value subject to a free entry condition for firms, as in Moen (1997). The optimal contract solves

$$U_0(a_0) = \max_{w_0, w_1, \lambda_{0w}, a_1, v_1} \lambda_{0w} V_1(a_0, a_1, w_0, w_1, v_1) + (1 - \lambda_{0w}) U_1(a_0) \tag{A.8}$$

subject to

$$\begin{aligned}
&\text{(IC-}a_1\text{)}: && u'(w_0 + a_0 - a_1) \geq \delta u'(b + a_1) + (1 - \delta)u'(w_1 + a_1) \\
&&& \quad + (1 - \delta)\left[\kappa \partial_a \lambda_{1w}(v_1, a_1)(v_1 - u(w_1 + a_1)) - \kappa\lambda_{1w}(v_1, a_1)u'(w_1 + a_1)\right] \\
&\text{(IC-}v\text{)}: && \partial_v \lambda_{1w}(v_1, a_1)(v_1 - u(w_1 + a_1)) + \lambda_{1w}(v_1, a_1) = 0 \\
&\text{(IC-}a_0\text{)}: && U_0(\hat{a}_0) \geq \max_{\hat{a_1} \geq 0, \hat{v}_1} \lambda_{0w} V_1(\hat{a}_0, \hat{a_1}, w_0, w_1, \hat{v}_1) + (1 - \lambda_{0w}) U_1(\hat{a}_0) \quad \forall \hat{a}_0 \neq a_0 \\
&\text{(BC)}: && a_1 \geq 0 \\
&\text{(FE)}: && -k + \lambda_{0f}\left(1 - w_0 + (1 - \delta)(1 - \kappa\lambda_{1w}(v_1, a_1))(1 - w_1)\right) = 0 \\
&(\mathcal{M}): && \lambda_{0f} = 1/(4\lambda_{0w})
\end{aligned}$$

$U_0(a_0)$ denotes the value achieved by an unemployed worker at $t = 0$ with initial asset a_0, a_1 denotes the savings decision of the worker conditional on finding a job and λ_{0w} denotes the job finding rate at $t = 0$. The first constraint (IC-a_1) is the Euler constraint that must be satisfied because the savings decision of employed workers a_1 is private information[18]. This constraint holds with equality if $a_1 > 0$. The second (IC-v) is the incentive compatibility constraint for search. The third (IC-a_0) is the incentive compatibility constraint that ensures that workers with assets $\hat{a}_0$ do not deviate and search in a market designed for workers with assets a_0 who will receive wages w_0, w_1. This is the new constraint that arises because the worker's initial assets are private information. A contract that satisfies this constraint ensures that workers have no incentives to

[18] In the 2-period model, labor markets at $t = 1$ can be expressed in terms of wage instead of values. With this formulation, it becomes obvious that equation (IC-a_1) is an Euler equation.

misreport their initial assets to firms. The fourth (BC) is the borrowing constraint of workers. The fifth constraint (FE) is the free entry condition of firms at $t = 0$. The last constraint ($\mathcal{M}$) is derived from the matching function. The equilibrium with observable a_0 in figure 6 solves (A.8) without (IC-a_0) but with (IC-a_1).

To solve this model, I rely on the algorithm proposed by Guerrieri et al. (2010). The first step is to identify the value of initial assets a_0 that no worker would falsely pretend to have. Since workers benefit by pretending to be wealthy in this model, this value is the minimum level of assets a worker can have at the borrowing constraint $a_0 = 0$. Because no worker would ever pretend to have $a_0 = 0$, we can solve the optimal contract (A.8) without the constraint (IC-a_0) and recover $U_0(0)$. We consider a small increase in assets to ϵ and solve for $U_0(\epsilon)$. In this problem, the global IC constraint (IC-a_0) can be replaced by the following local IC constraint: the contract should be designed such that workers with initial asset $a_0 = 0$ do not want to pretend to have assets ϵ. We then consider the value of initial assets $2 \times \epsilon$ and proceed iteratively to compute $U_0(a_0)$ for all a_0 on a grid. That is, at each grid point a_0, the local constraint is that workers with assets $a_0 - \epsilon$ do not want to pretend to have a_0. I then check ex-post numerically that the local IC constraints are indeed sufficient for global incentive compatibility. The set of contracts that we construct in this way ensures that no worker would misreport his initial assets. In the numerical implementation, I consider $\epsilon = 0.002$.